

Feeling sight



2021 HSC Design and Technology
Oxley College

Contents

Project Proposal & Management	3
Identification and Exploration of need	4
Areas of investigation	7
Project Constraints	8
Criteria for Evaluation	9
Test analysis	10
Project management	11
Action and finance	13
Motivation	16
Ongoing evaluation	17
Project	18
Development	18
Generation of ideas	19
Consideration of factors effecting design	20
Research into existing designs	23
Research and experimentation	25
Application of conclusions	51
Identification and Justification of ideas and re- sources used	53
Evidence and application of practical skills to produce a quality project	55
Ongoing evaluation	70
Degree of difference	72
Project Evaluation	73
Analysis and evaluation of functional and aesthetic aspects of the design	74
Final evaluation with respect to the projects impact on the individual, society and the environment	76
Relationship of the final product, system or envi- ronment to the proposal	78
Final Product to End User	79
Acknowledgements	80

Project Proposal & Management



01

Identification and Exploration of need

Brief

Develop a system that will allow motorcyclists to know exactly where other vehicles are around them to increase safety and reduce accidents

Aim

Help motorcyclists know exactly where they are in space and where other vehicles and dangerous obstacles are without needing to see them. This will help riders avoid collisions and increase safety on road and track.

Exploration of need

Serious road accidents in 2019, which was a year with far reduced motorcycle injuries, still had bikes accounting for 37% of all accidents. But this statistic seems not too bad until it is realised that motorcyclists only account for 5.7% of road users. **Motorcyclists are 30 times as likely to be killed in an accident than a car occupant.** These statistics beg the

question: why are motorcyclists so much more likely to be killed in an accident than car occupants? The answer lies in the fact that motorcyclists are not protected by a metal cage like car occupants are. They are also more likely to be wearing helmets, which can reduce the severity of head injuries, but they are still more vulnerable than car occupants.

Current safety precautions

The safety precautions taken for motorcyclists currently are protective gear (e.g. helmet, pads, etc), mirrors, lights and riders intuition. This set up has been adequate and has helped many riders survive accidents but the statistics show that there is large room for improvement.

Leading accident causes

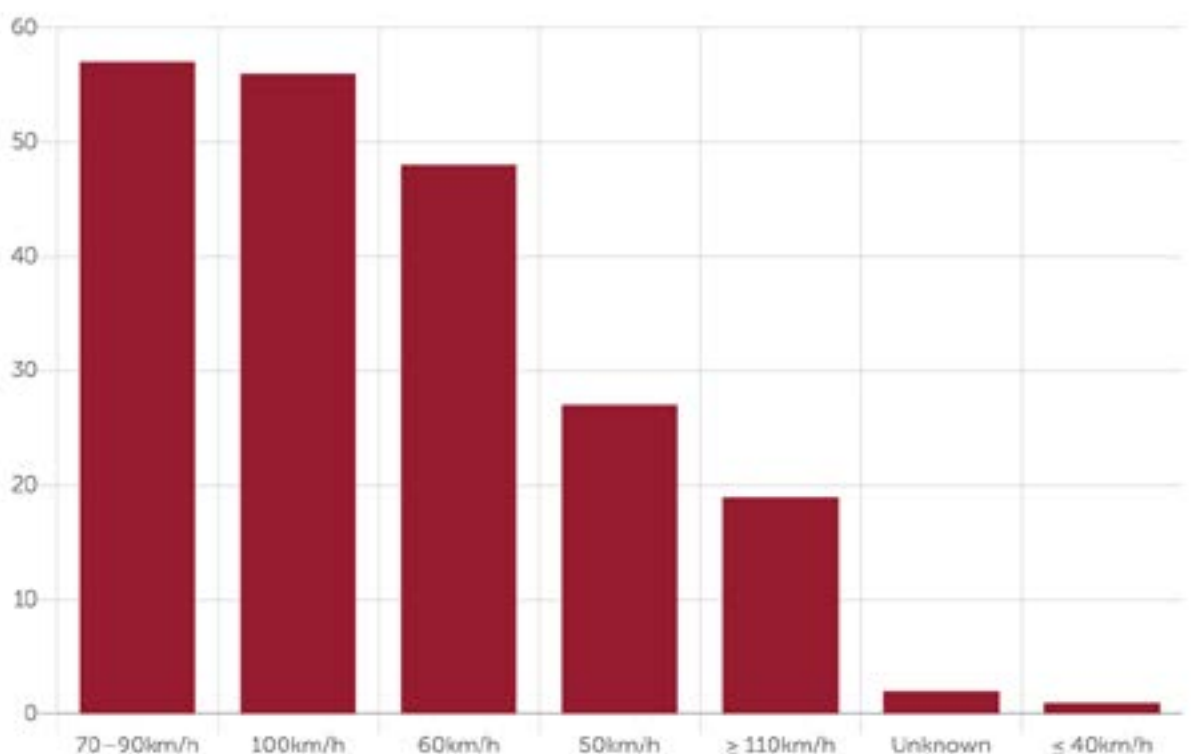
There are a few leading causes for accidents involving motorcycles these are;

1. Improper or unsafe lane changes
2. Speeding or driving too fast for conditions
3. Intoxication
4. Lane splitting
5. Car doors opening
6. Rear end collisions
7. Inexperience
8. Right-turn intersection
9. Road hazards
10. Defective equipment

Film making. This information points to a solution that helps the rider with making

more sensible decisions in moments when precautions like head checks are forgotten.

Death
by
speed
limit

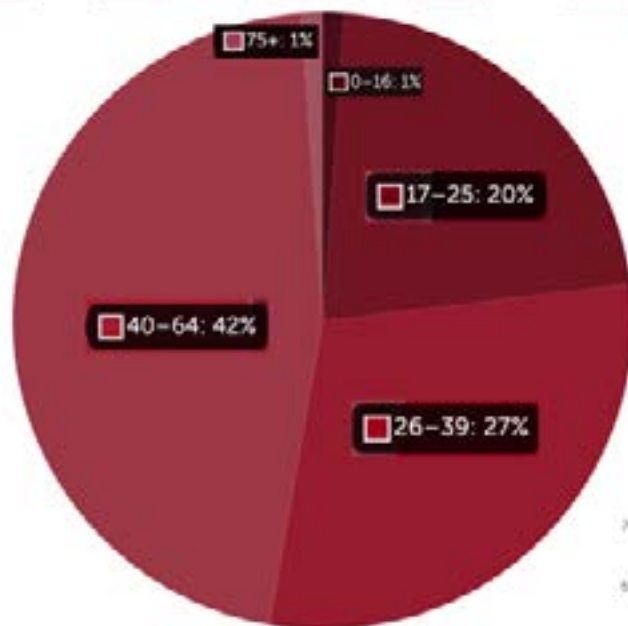


Target market

Target market research

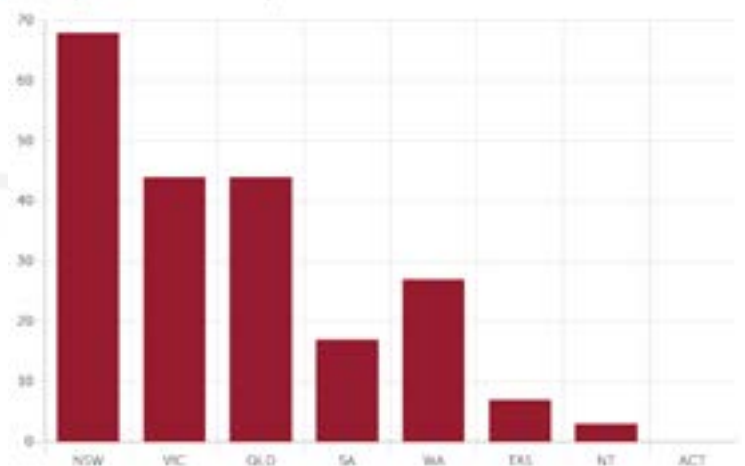
- Interviews with motorcycle riders that ride on the road.
- Interviews with emergency service people who attend crashed to do with motorcycles.

0-16 17-25 26-39 40-64 75+



Severe injury
by age

Deaths by
state



Target Market

Age: 40-64

Gender: Male

State: NSW

Occupation: Short distance commuter (middle income)

Choice of settlement: City/Suburban

This information is obtained by researching what age group and gender has the most registered motorcycles on the roads of Australia. Then each states registered motorcycle numbers are tallied and NSW turned out to have the most registered motorcycles.

Feeling sight

Areas of investigation

These questions are what will be asked to current licenced motorcycle riders to gauge what the target design should be.

Target market Questions

- What do you consider to be the most dangerous part about riding?
 - What would you recommend as the safest thing to use, wear or do when riding?
 - Are there any unexpected safety considerations that should be considered when riding?
 - Do you think the roads are safe for motorcycles?
 - How should someone try and ride to be as safe as possible?
- □ Phone number _____
Address _____
- Physics and where vehicles are placed around you?

Area of Investigation	What needs to be Investigated	Method of research
Target Market	Motorcycle riders on roads and tracks	• Interviews • Articles and videos
Materials	Elastic	• Talk to textile professionals
	Silicone	• Talk to Industrial designers
	Lightweight Fabric	• Talk to textile professionals
Tools	Sewing machine	• Use prior knowledge • YouTube
	Mould	• Talk to Industrial Designers
	3D Printer	• Use prior knowledge • Work with 3D printer technician
	Lidar/Cameras	• Talk to professionals in this field • Research online
	Haptic Motors	• Research online • Use research
Techniques	3D Design	• Use prior knowledge • Use forums
	Arduino	• Forums • YouTube • Prior knowledge
	Raspberry Pi	• Forums • YouTube • Prior knowledge

Project Constraints

Time	The overarching limitation of this project is time. From the point that this project commences to the marking date is enough time to be able to create a nicely working prototype but will not be enough time to create a working product ready for production and sale
Cost	Cost is a major factor that is an unchangeable boundary. This project will have to be kept to a cost as low as possible and will be based on parts that may not be the highest of quality to start with but at the end of the project will be substituted for good quality parts.
Knowledge	Knowledge is a limitation but knowledge can be obtained from large amounts of research and trail and error. The amount of knowledge that can be obtained in a short amount of time is a limitation so finding the correct pieces of information first up is essential
Open-source nature of technology	Luckily today many technologies are open-source but there are a few that restrict access to the public. This could potentially be a large issue that will have to be dealt with when it occurs.
Electrical/Computing certification	I am not a certified electrical or computing professional which limits what I am allowed to create as a product. I can not for example connect any type of component to mains power or can not access computing services and technology which will only be distributed to professionals.
Biology	A boundary which will have to be surpasses is that I have very minimal critical knowledge to do with biology and the humans neural network. I can will do large amounts of research which will help me understand and achieve my desired outcome, but I will not know all the contributing factors as I am not a professional.

Criteria for Evaluation

	Function	Aesthetics/Ergonomics
1	Will provide rider with information about vehicles around them	User is able to feel the motors even with an engine running
2	Will Provide user with information about their surroundings and any other analytics	Discrete in clothing
3	Battery pack and processing unit able to be stored safely	Does not create bulges in fabric
4	Has battery life that will last at least a full day of activity or have changeable batteries.	Comfortable
5	Connect to power source from the Motorcycle	Fits body and sits in same place each use
6	Be able to measure corners and how fast someone should go into them	Can be adjusted for different users

It is crucial that this suit informs user of where vehicles are around them because this is the purpose of this suit and if it were to not work it could be risk-

The user must be able to feel the haptic pods operating as this is the way of the device communicating with the user and if this communication is forfeited

Test analysis

Function

Rider will test device on road to see if it enables them to navigate roads with increased safety.

The ability for the device to extract live data from the bike will be tested by running device alongside running bike. And information coming into the processing unit will be measured.

The device will be tested and will be made sure the battery will not run out and leave the rider a sense down during the day and that the battery pack and processing unit is safely stored which will mean it is water resistant and durable to withstand impacts.

This will require hardwiring to the bikes power supply. This may not be possible but could be a later modification.

Test the ability of cameras to measure the angle and severity of corners and to translate that into motor vibrations.

Aesthetics/Ergonomics

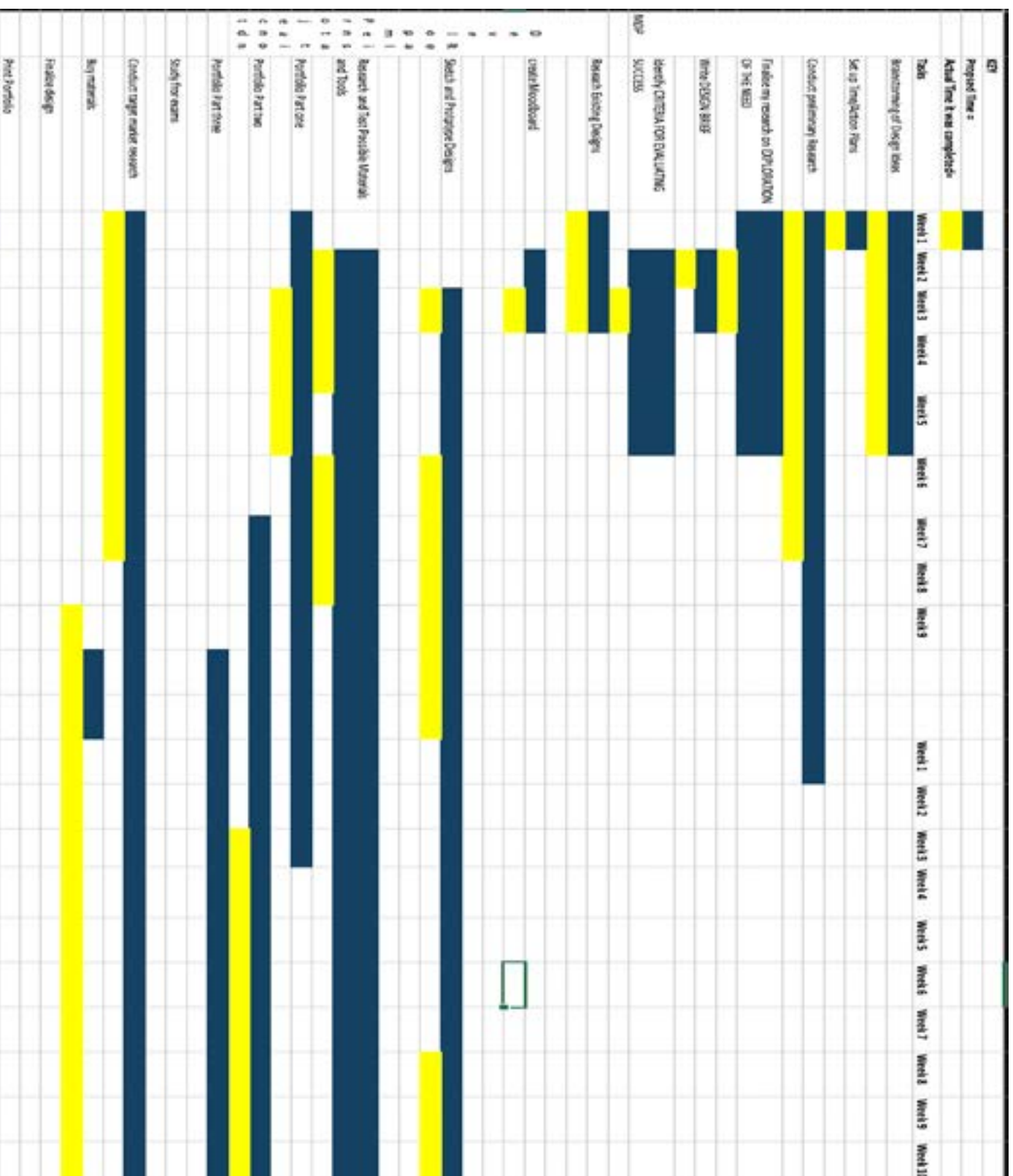
The rider will have to ride with engine running and on different types of roads to make sure that the motors are always able to be felt.

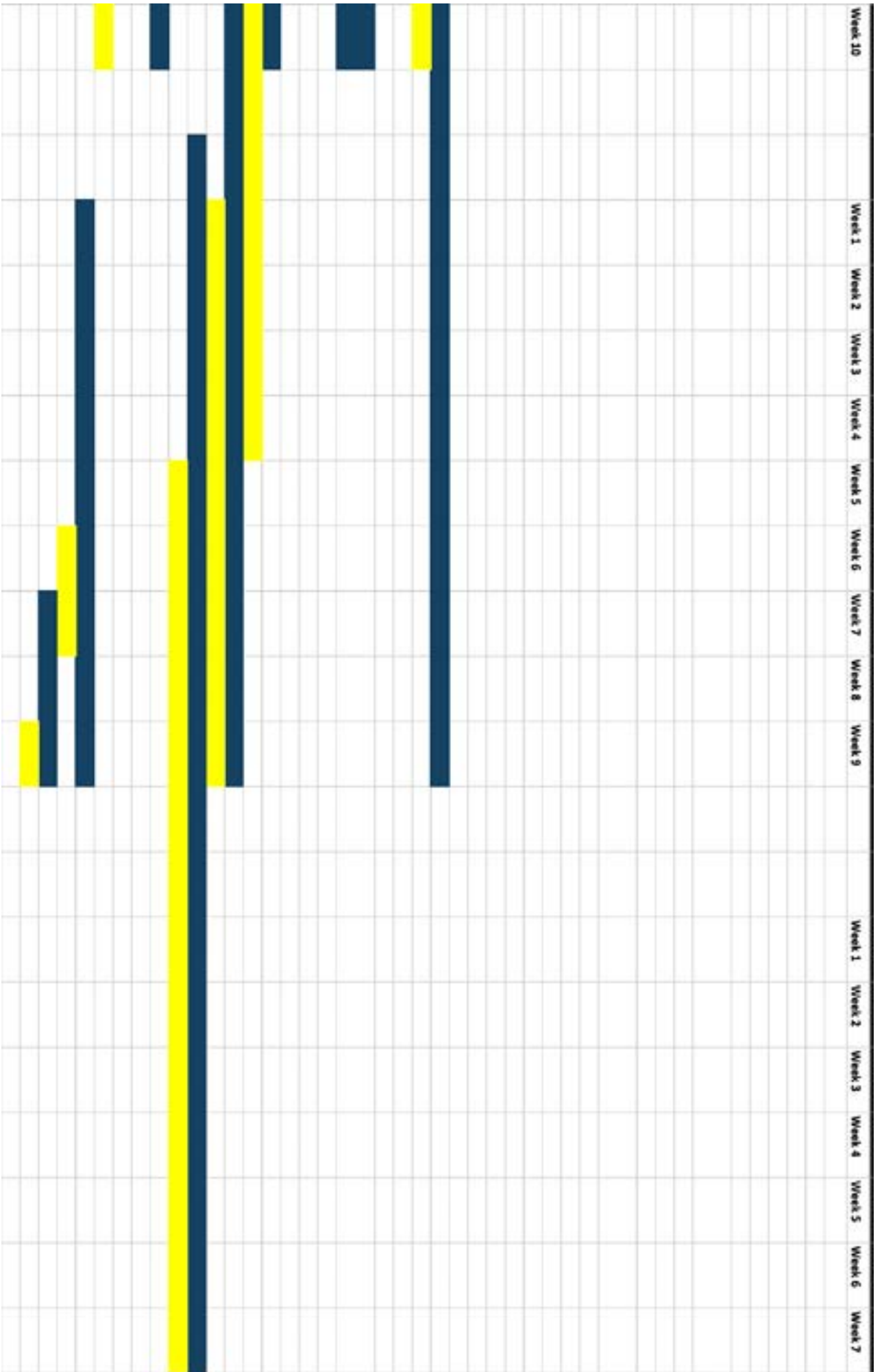
This will require the device to be worn with all types of biking clothing to see if it is visible underneath the clothes.

Different riders will need to try this device on to see if it is comfortable and is able to be adjusted to fit them perfectly easily.

Project management

Time Plan





Action and finance

Resources	Predicted cost	Actual cost
Haptic motors	\$2.57 <u>each</u> x20 = 51.40	\$19.60 x 2 = \$39.2 For 30 motors
Arduino Mega	\$27.45	\$27.45
Raspberry Pi	\$66.92	\$66.92 But already had one (didn't use for project)
Camera	\$34.99 x 4 = \$139.96	\$34.99 x 2 = 69.98 (<u>did</u> not end up using them)
Elastic	\$5 per 2m x2 = <u>\$10</u>	\$17
Silicone	\$27.10 per 1kg	N/A
Mould (3D print material)	\$30.92 per 1kg	School Provided
Synthetic Sports Mesh	Did not predict	\$40
Neoprene	Did not predict	\$23.1 x 5 = \$115.5
Internet cable	Did not predict	\$6.5 For 5m
Multipin Nylon Plug	Did not predict	\$15.9
Switch	Did not predict	\$15.9
Wires and Bread board	Did not predict	\$8.7
LCD	Did not predict	\$14.85
Lidar	Did not predict	\$42.5 x 3 = \$127
Ultrasonic sensor	Did not predict	\$3.85 x 2 = \$7.7
L293D Motor Driver	Did not predict	\$9.3
Shipping, handling and GST	Did not predict	19.94+17.63+2.87+12.38 = \$52.82
Total	\$353.75	\$567.8

Finance Evaluation

From the onset of this project I aimed to complete it as a budget build, which for me was going to be around \$250. I resulted in spending far more than what I had hoped. This is because each step of the way I would have more ideas and want this project to succeed as much as possible. So the reason I kept adding things to the build was because I wanted to make this project into something that is of higher quality than it would have been. If I were to remake this project I believe I could make it for a lot cheaper as I have now learnt a lot about where I can bulk buy electronic parts and where if Covid was over I could source cheaper and potentially better quality neoprene.



Reciepts



Item	SKU	Qty	Subtotal
HC-SR04 Ultrasonic Module Distance Measuring Sensor	CE05112	2	\$7.70
TF-Luna LIDAR Module - Short-Range Distance Sensor	SS101990656	2	\$85.90
	Subtotal		\$93.60
	Shipping & Handling		\$19.89
	Included GST		\$10.32
	Grand Total		\$113.49



Motivation

The reason for which I wish to do this project is because I have a deep interest in prosthetic senses. Ever since I had done some external research on re-mapping brains to learn new senses I have wanted to make a suit that helped a group of people either have increased ease or safety with their life. What came to mind was to make a suit for blind people, but due to not being able to access many blind people and there already being products like this being developed I decided against it. My other idea was for motorcycle riders and car drivers to be able to sense where obstacles and other vehicles are around them. As I ride motorcycles I decided that I would like to make my project for motorcycles. This is because I have started to realise some of the severe dangers that motorcycle riders face when riding especially on road. Most people will know of someone that has been in a motorcycle accident whether they are family or friend or a friend of a friend. Motorcycle accidents can be very devastating but I hope that I am able to contribute something which will help lessen the chance of getting into an accident.

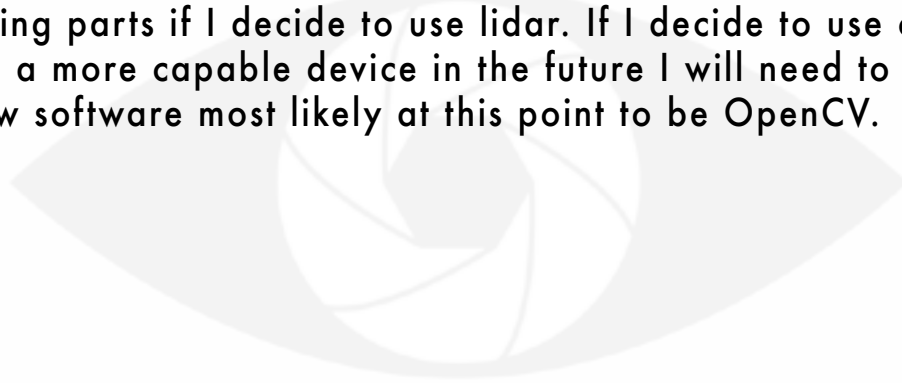


Ongoing evaluation

This project so far has been running smoothly. There have been no major issues that have suddenly occurred. I have changed the need for my design I originally was thinking of making a device to help blind people move access their environment with greater ease but instead I changed the need back to my original concept which is for motorcycles. This is due to not having access to blind people to do constant tests with and that there are other ideas and my project would be capable of doing in such a small amount of time.

My project is moving along as expected, except for making a logo. I have tried thinking up a good logo idea but there have not been any suitable ideas just yet.

I have put a large amount of research into different areas which is influencing my decisions to do with the first prototype of this project and have already written the basis for the code which should be one of the most time-consuming parts if I decide to use lidar. If I decide to use cameras which will result in a more capable device in the future I will need to learn how to program new software most likely at this point to be OpenCV.



Project Development



Generation of ideas

The reason why I am realising this design is because I am a motorcycle rider and know how dangerous riding on the road can be. We all know someone who rides and if they are involved in a crash they are 30 times more likely to die. So to help reduce this statistic to this major issue is to find out why people are getting into these crashes, and the over whelming reason was because of improper or unsafe lane changes and the other reasons which have nothing to do with the riders negligent decision making like riding drunk was, lane splitting and rear end collisions. All these accident catalysts are causes that could happen to the most experiences riders. I had to figure out a way to make the rider more aware of what is going on with the road behind them.

To do this I came up with the idea of having haptic feedback pods on the back of the seat. Haptic feedback it is using a dormant sense to transfer information to the rider. There is no haptic feedback suits in use for any type of motorsport or civilian travel on the market or in public development today. Therefore I have created this safety suit it is a totally different type of safety gear. It does not protect the rider when they are in a crash, but it helps prevent a crash. This device instead of transferring the information to the motorcycles system transfers it to the human. The reason why this information can never be transferred to the actual motorcycle system is first of all the bike does not have control over steering and usually does not have control over braking, and even if it did it would unbalance the rider and potentially cause more accidents than it would save.

Consideration of factors effecting design

Ergonomics

- Comfortable for user
- If it is not comfortable it will not be used when needed
- This Suit needs to be able to be a solid but comfortable builds

Safety

- Project is to increased ease and safety on the roads and tracks, motor-cycling can be very dangerous but this can be lessened by taking precautions such as wearing a helmet and using devices such as this.
- This device is not required to be something that saves a life by physically protecting in case of an accident but is rather a device to prevent accidents.

Aesthetics

- Device will not be used in all situations if it affects the persons appearance

□ □ Major aim is to feel light and comfortable for people to get over to be worn.

Cost

- Needs to cost as little as possible so retail will be economical and anyone who may need it can buy
- This needs to be built with quality but also in a replicable less financially straining way.

Accessibility

- This device needs to be accessible to anyone who may need it.

Considerations of design factors relevant to the major design project

Finance

Finance is a very important factor to consider when designing a product. This product is one which can very easily suddenly start costing a lot of money. Electronics cannot be made from the ground up they have to be sourced from somewhere else and then integrated together. This is where small and seemingly insignificant but essential parts start costing a lot of money. And if a cheap electronic part is used they often don't work properly or only work in a perfect environment, for example they may not have static protectors which makes handling them incredibly dangerous for the actual pieces. The other cost which has to be carefully factored, is the testing of different materials to make sure they do exactly what you want. I have found that testing different materials for the suit added up more than I had planned for. The final costs that I have had to consider is the final purchase of all materials. To lower this cost I bulk bought different things like the motor controller chips which are usually \$5-\$7 a piece but I was able to get 6 for \$12. Unfortunately I was not able to do the same for the neoprene which turned out to be the most expensive part of the project. Due to Covid-19 the only place I could source neoprene from in my area was Spotlight. As this project was to build a whole new type of device I was not able to recycle much this meant I had to buy everything new.

Time

As this project has only had under a year to develop it has not yet hit its full potential. There are many more full prototypes to be had and prototypes which would cost far more than what has been produced currently. With the given time this project will have to have only the essential work put into it to make it work and look good. This time frame does not give enough time for this project to be made into a sellable product. But in this time frame the purpose, function and aesthetics of this project can be demonstrated.

Ergonomics

The function of this project is to ergonomically work well and transfer information to a user through feeling. This project is required to feel comfortable and tight fitting to the body so the haptic motors can be felt when riding a motorcycle. This suit and haptic pods have to have a seamless design to not effect the other gear's functions and ergonomics.

Aesthetics

The two aesthetic pieces of the design is the fender box and the suit. The fender box has to match the aerodynamic look that motorcycles have. This will be achieved by gaining influence from aerodynamic designs such as aeroplane wings and racing spoilers. This design can look like a racing spoiler because it is a widely accepted design that looks good on motor vehicles. So a discrete spoiler on the rear of the motor cycle would be a design not to uncomfortable to the eye. The suit has to be making no statements at all, it should be very discrete and not make normal clothing fall in the wrong way. The suit needs to not effect the look and useability of normal bike gear. The one issues with aesthetics that will not be able to be fixed in this project just now is that the suit and the fender box need to be physically wired together. This wire will not look great and will effect the overall look of the device. But in future prototypes this wire can be eliminated by making the suit wireless which would make the useability of the device far better.

Safety

The main purpose of this device is to make motorcycling on the road safer. There are many motorcycling accidents and a way to help eliminate some of the dangerous aspects of motorcycles that cause these accidents is to tell the rider where other vehicles are behind them. The main danger that this device could have to a rider is that the haptic pods are close to the body and in the event they could cause injury to the user. So to have no injury toward the user the suit is made from 3 layers of neoprene which provide padding and the haptic pods designed to not have one point in which a large amount of pressure is applied on the body. The haptic pods are flat with rounded edges which mean injury from them will not occur.

Societal

Societal design factors have to be heavily considered because this ties in with hitting the target market. The societal design factor for this project shows that it is required to have a device that does not detract any of the enjoyment of riding motorcycles and that it does not become a device that is considered pointless. But more than that this project will hopefully save lives creating a better, safer society for all road users. In a world that is becoming more and more autonomous for four wheeled vehicles it is essential that motorcycles do not get overlooked as so many people still use them as a means to travel places.

Research into existing designs

Classon helmet by Brooklyness



Positive

Great for cyclists
Brings signals that most road vehicles have to cyclists

Negative

Does not tell user exactly what the vehicles in the blind spot are doing.

Evaluation

This helmet is great for making cycling travel safer and brings in signals that are universally understood but it does not tell the user what the vehicles are doing in the lanes behind.

Hardlight VR Suit



Positive

The suit can provide the user with full virtual reality making them feel as though they are in the Generated world through haptic sensors

Negative

Many of these parts are difficult to manufacture which was a reason why they failed as a company

Evaluation

This suit has some very good concepts, but failed as it was too difficult to manufacture.

Existing designs

Suit that helps deaf people understand speech

The limitation with this suit is that it cannot be applied to bikes without significant changes to sensors and processing and motors are not placed at strategic places along the body for use with motorcycles.



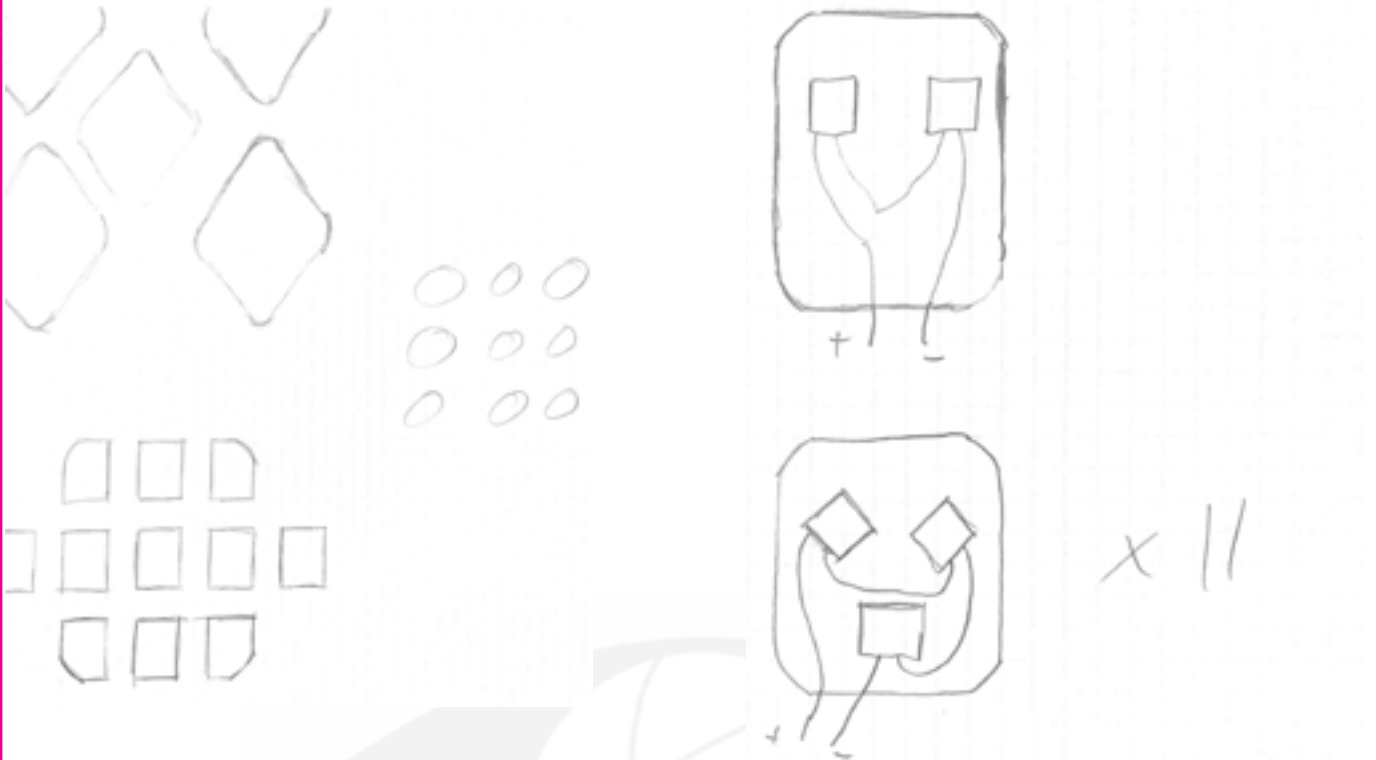
Rear mirrors, and blind spot detection

Motorcycle blind spot detection (BSI) has advanced greatly since then. Some other rear view mirrors have blind spot detection, this is a slight step up but no information other than the fact that a vehicle sits in the riders blind spot is available.



Research and experimentation

Haptic motor pod development



Aim

To make a haptic pod that holds haptic motors and create the most vibration as possible

Method

Develop different types variations of the haptic pod. Compare side to side

Results

When a third motor is put in and all 3 motors are put at 45 degrees from each other the haptic pods work better



First Prototype

Final Prototype

Improvements made

- Moved screw slots out to edges to allow for more space to fit motors
- Decreased size of box
- Added third haptic motor slot at 45 degrees from each other compared to being parallel. This creates a vibrations in more directions

Finding fabrics

Aim

To find a fabric that will support haptic motor pods and its performance is not effected by sweat or excessive movement.

Method

To look around at shops to find many different types of fabrics and descriptions of how they perform when in a excersice movement situa-tion.

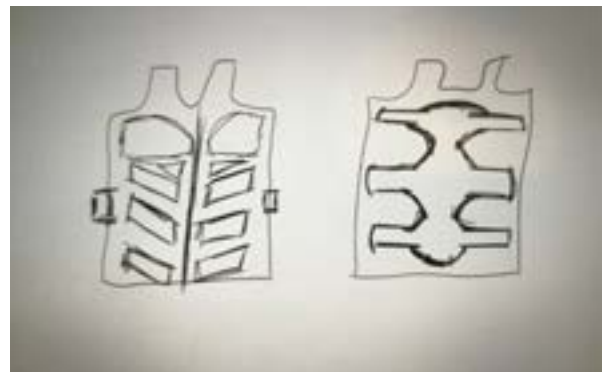
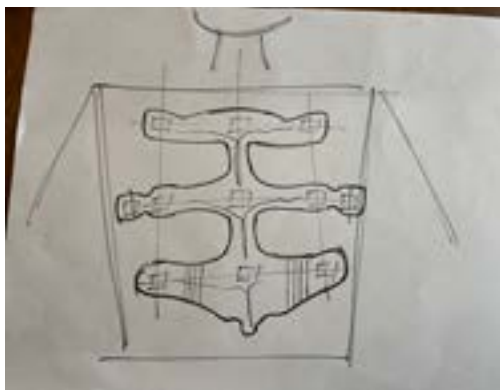
Results

Found that neoprene is an incredibly good material because of the support it provides while being moveable and uneffected by moisture. And sports mesh poliester fabric is verry good for being uneffected by mois-ture and has multidirectional stretch



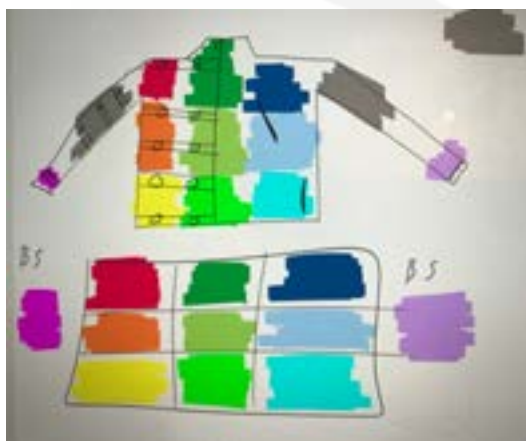
Looking at different fabrics that are used in the textile industry and coming to the conclusion that I will use neoprene and the sweat wicking sport fabric used in many peices of sports gear.

Suit design and preparation



Trialing what designs are aesthetically pleasing and would function correctly. This is too much like a superhero costume.

Positive	Negative	Interesting
Satisfies all the requirements for the suit to hold haptic motors.	Was thinking of making frame piece out of soft rubber and molding with heat. This will be too difficult as have never done something like that before.	This would have been an interesting design to see if it was a comfortable design.



Concept of suit. Decided to have 3 columns of haptic pods to represent the traffic lanes around a rider. There are also a two blind spot haptic pods. And potential for having vehicle identification Pods.

Positive	Negative	Interesting
Easily can be understood by a person new to the technology.	Having vehicle identification Pods may be too complex as a first prototype suit.	It would be interesting to have as many different Pods throughout the suit To convey different information.

Aim

To form a shape that sits naturally on the human body and provides function by assisting in feedback of the Haptic pods.

Method

cut, I will test to see if they fit me. So the suit functions to the best it can. I tested putting haptic motors on a range of peoples backs to see where they felt them most. The average of these results are where the Haptic pods will be positioned.





Results

The Cardboard suit sits well on my body and also my fathers which is a good result. The Average area for where people most felt the Haptic Pods in now incorporated in the suit. These results have interestingly fallen around known pressure points of the back.

Conclusion

I will use this cardboard mock-up as a template for cutting the neoprene.

Aim

Make the suit push the Haptic Pods against the body to ensure full contact so information can be seamlessly transferred to the user.

Method

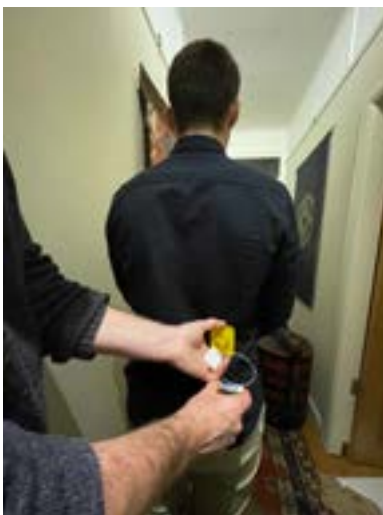
Brainstorm and experiment what types of techniques would provide pressure of the haptic Pods against the body.

Results

The only way to apply pressure to a certain part of the body from a wearable that does not protrude far from the body is to tighten it. But I do not want a suit that has to be incredibly tight to work. The only way to achieve this is to have straps pull on either side of the section that needs to be tight. This allows the suit to not need to be so tight for use. With the use of correctly tensioned elastic, this allows for movement and more comfortable, unrestricted use while keeping pressure on the essential spots. The downward angle of the straps after being wrapped around the body, end up horizontal and parallel which is what is wanted to provide adequate pressure.

Conclusion

To keep pressure on the haptic pods there will be two straps extruding from the bottom two rows. These straps will be angled down to provide so after they have been wrapped around the body they will sit neatly. And the straps will have elastic at some point through them so the suit can be moveable and more comfortable than just neoprene.



Testing haptic pod on back to find most sensitive areas

Fendor Box - Research

When coming up with a fender mount box to hold all the electronics I had to firstly consider the size of the electronics and make a box that they could fit in. Next, I had to consider the shape of the fender and work out the basic dimensions I was working with for the fender. After measuring up both of these aspects I then designed the fender box in a way that looks aesthetically pleasing and also has the function of having an LCD screen and various control buttons and port to plug in the haptic pods.

Aim

Calculate the average fender shape of most common brands of motorcycles. Work out the maximum volume that can be created with the base dimensions of the motorcycle fender. Measure maximum length, maximum width and the curvature of the fender in both x and y directions.

Method

Measure as many motorcycle fenders as can be found in a large range of brands. These bikes will be found mainly at motorcycle retailers. Use straight edged metal rulers and right angel set square to work get accurate measurements.



Results

The **Husqvarna FE 500** dirt bike had the **smallest area to fit mount a box**. This will be the maximum base area that the box will have to fit, because if the box fits in this area it will fit all other bikes measured and most others in production. All of the motorcycles measured had **no** curvature along **length** of the **fender**, this is due to modern day popular styling which is currently promoting sharp unrounded designs. There are motorcycles that have curved fenders but these are **filmerhejkibdkkabejkgjlddkMakkhma** **fender width curvature** was calculated to be **35mm below the apex**. The average was taken so that the box will fit the majority of motorcycles while looking flush. And by using 3M foam double sided tape any gap even on perfectly straight fender would be evened out and would look normal and be mounted well.

Conclusion

Conclusion
This haptic pod will be designed to fit the Husqvarna FE 500 rear fender with a curvature of 35mm below apex. Whilst having enough volume to contain all electronics.

Initial ideas for fendor box

Aim

Design a box that holds all the electronics required to operate this project and make a box that will fit within the maximum measurements that have been determined by measuring motorcycles in the previous investigation.

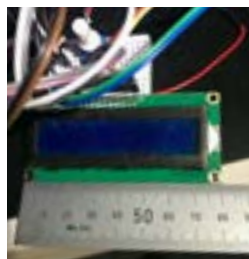
Method

- 1) Take inspiration from aerodynamic designs such as car rear spoilers and aeroplane wings
- 2) Identify what type of box design is aesthetically pleasing on the back of a motorcycle while being slightly discreet. Do this by making concept drawings
- 3) And once a good look is identified make a rough box design out cardboard and tests how it looks on the bikes and how well it fits the electronics.
- 4) Then design this box on Fusion 360 making sure every measurement is perfect to the millimetre. Make sure that all the electronics are measured accurately and design mounts and holes for where the electronics will sit.
- 5) Then 3D print



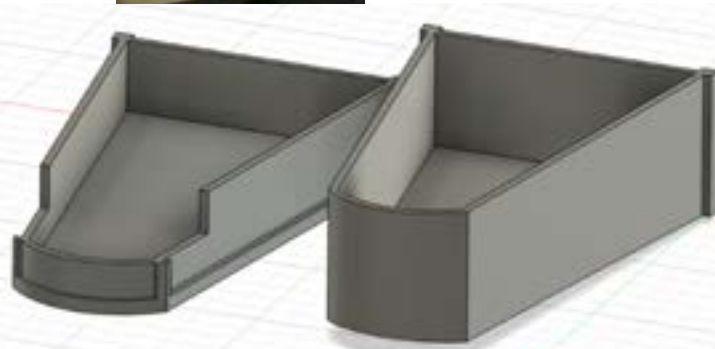
Testing what shape may look best by drawing

Measuring Electronics



Testing which way the shape is oriented to get the most aesthetically pleasing look

Lip designed into frame so the lid has rails to guide it and the lip also holds the lid on.



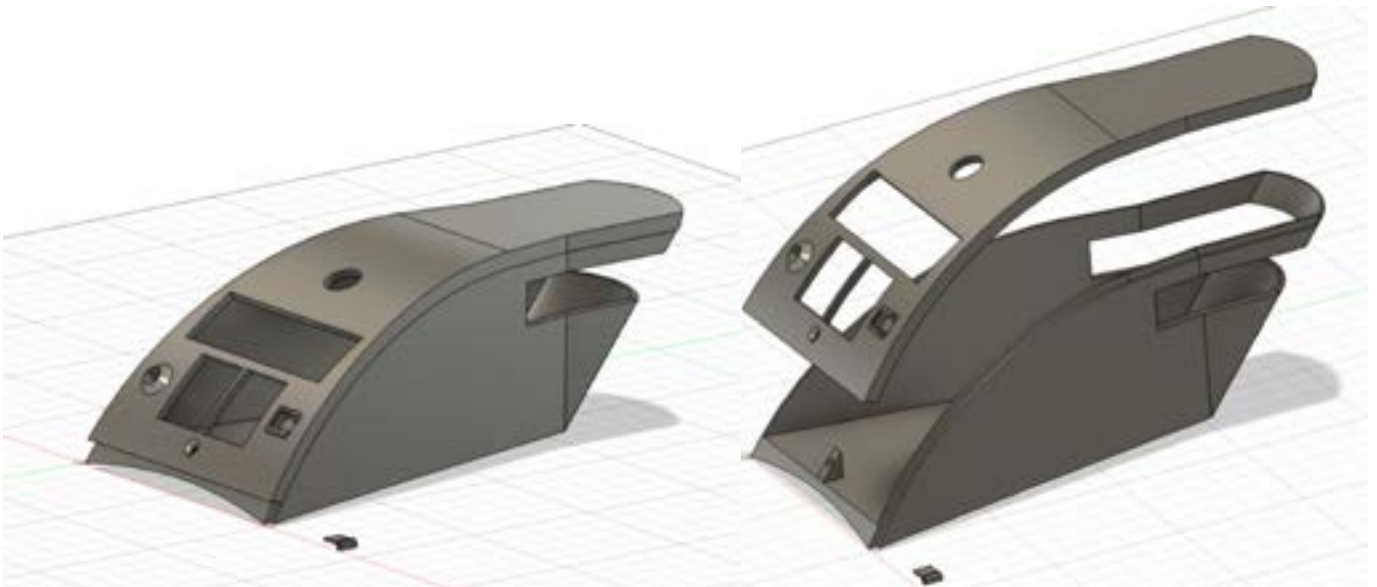
First design which was discarded in the end

Results

The designed box developed well through all the steps to realise its creation. Features that the box has is natural aerodynamic shape one similar to that of a rear spoiler on cars. It also has holes designed to hold haptic pod connection port, mode button, LCD screen and display brightness, ON/OFF switch and most importantly 180 degrees of open vision for the sensors.

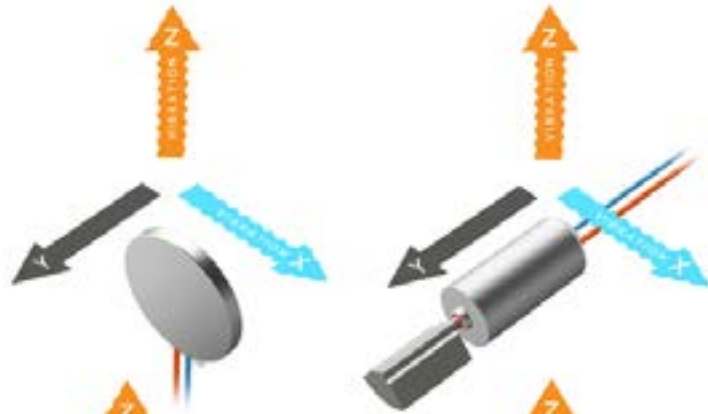
Conclusion

This box will fit most bikes and will hold all the electronics well.

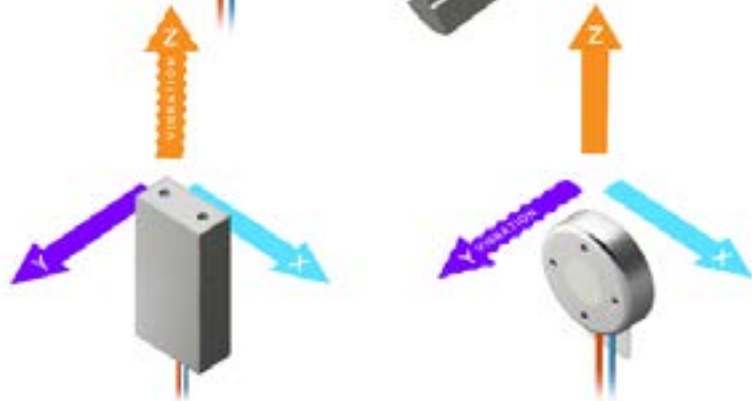


Haptic Motor Choices

ERMs



LRAs



Linear vibration motor (LRA)

This type of haptic motor has strong vibration along one axis. This makes this type of motor is ideal in uses such as in mobile phones. This is because developing phones are utilising unidirectional motors to provide fake feelings of buttons. For example the iPhone 7 and 8 home button.

Eccentric rotating mass (ERM)

This type of haptic motor has strong vibration along two axis. This makes this type of motor is ideal in uses such as this motorcycle suit. So it can be angled in different ways and still be felt.

Aim

To find which motor is best at transmitting vibrations when mounted in a plastic case.

Method

Have both different types of motors and test on multiple people which one they can feel most.

Results

As expected the ERM Haptic Motors worked best. They were able to transmit more energy through the plastic. The LRA also had a major downfall that they would have to be turned on their axis to generate acceptable vibration, this detracted from space in the case.

Testing Code

Aim

Make program that utilises lidar capability

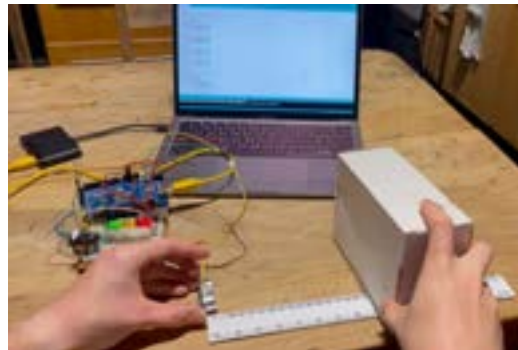
Method

Write code and test at different distances

Results

This code works well without faults as long as the device is wired correctly

```
Lidar_Test
#include <SoftwareSerial.h>
#define SoftwareSerial Serial1(18,19);
int dist;
int strength;
float temprature;
int check;
int i;
int uart[9];
const int HEADER=0x59;
void setup() {
  Serial.begin(9600);
  Serial1.begin(115200);
}
void loop() {
  if (Serial1.available()) {
    Serial.println(" ");
    if(Serial1.read() == HEADER) {
      uart[0]=HEADER;
      if (Serial1.read() == HEADER) {
        uart[1] = HEADER;
        for (i = 2; i < 9; i++) {
          uart[i] = Serial1.read();
        }
        check = uart[0] + uart[1] + uart[2] + uart[3] + uart[4] + uart[5] + uart[6] + uart[7];
        if (uart[8] == (check & 0xff)){
          dist = uart[2] + uart[3] * 256;
          strength = uart[4] + uart[5] * 256;
          temprature = uart[6] + uart[7] * 256;
          temprature = temprature/8 - 256;
          Serial.println("dist = ");
          Serial.print(dist);
        }
      }
    }
  }
  delay(100);
}
```



This program is testing an LCD Screen by displaying changing text on the screen

```
randomtestthings
#include <Servo.h>
Servo servo;

void setup()
{
  servo.attach(8);
  servo.write(0);
  delay(5000);
  Serial.begin(9600);
}

void loop()
{
  Serial.print(" 90 ");
  servo.write(90);{
    delay(2000);
  }

  Serial.print(" 0 ");
  servo.write(0);{
    delay(1000);
  }
}
```

This program works the Servo (A Geared motor) and writes into a serial port (Port on computer) the angle which the servo is at.

```
LCDTesting
#include <LiquidCrystal.h>
const int rs = 12, en = 11, d0 = 5, d1 = 4, d2 = 3, d3 = 2;
LiquidCrystal lcd(rs, en, d0, d1, d2, d3);

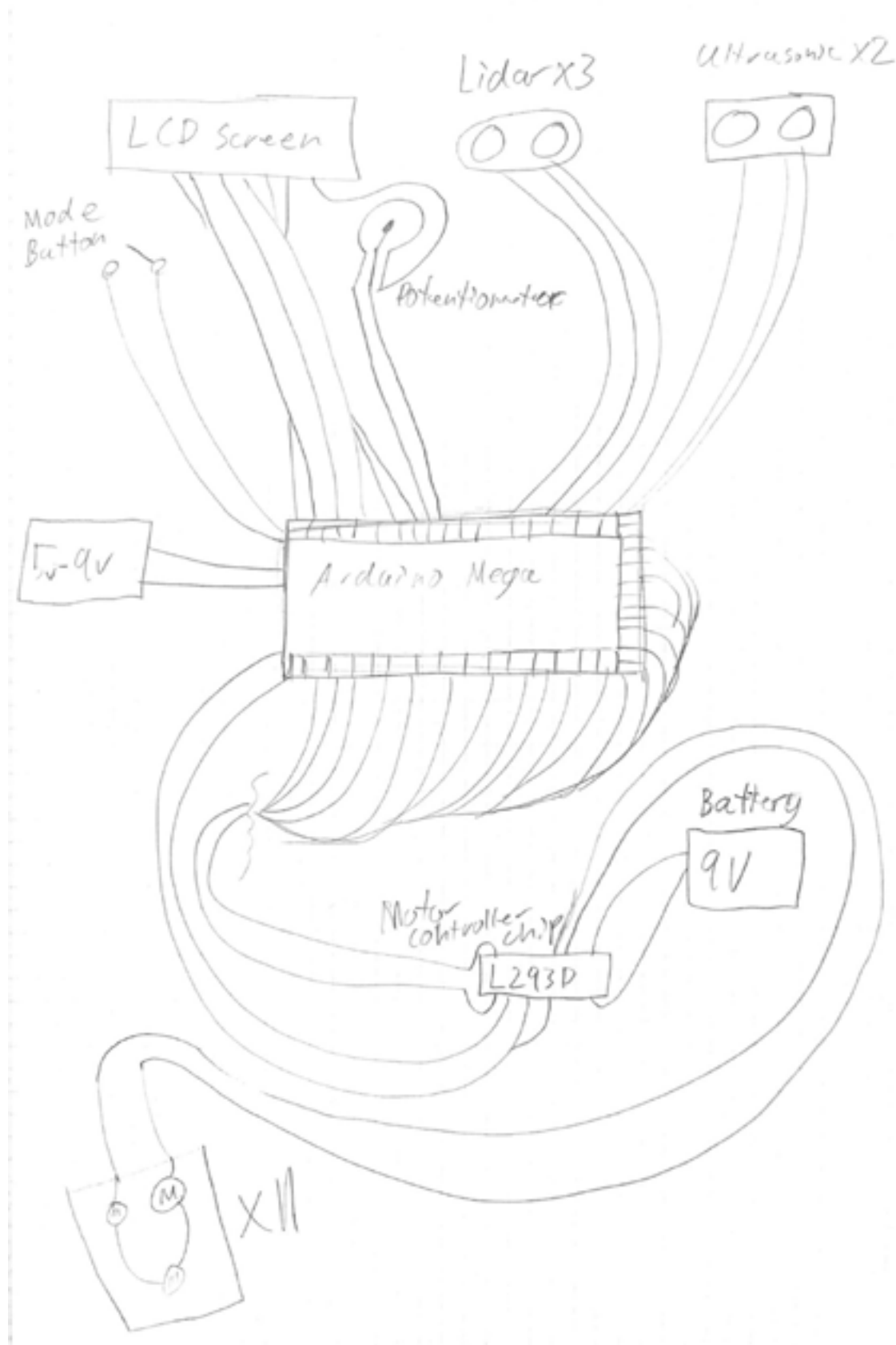
void setup() {
  lcd.begin(16, 2);
}

void loop() {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("WOP Testing");
  delay(1000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("10");
  delay(1000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("9");
  delay(1000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("8");
  delay(1000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("7");
  delay(1000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("6");
  delay(1000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("5");
  delay(1000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("4");
  delay(1000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("3");
  delay(1000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("2");
  delay(1000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("1");
  delay(1000);
}
```



Full Wiring Digram

Wiring the Arduino takes a lot of planning and testing. This project had many external components that had to be accounted for and have enough power going to them. To work out how much power each component need requires equations and allowing a margin of error because no electrical device will work the exact way you believe it to.



Testing and development of Haptic pod wiring

When components are attached to an Arduino there are many small additional physical tricks that are required to operate everything synchronously. Having everything work first go is an incredibly rare occurrence. So testing all these components separately is essential as sometimes one wire may be connected to a wrong port and this can effect a component on the other end of the board.

Test 1

Aim

Make sure all chips operate correctly

Method

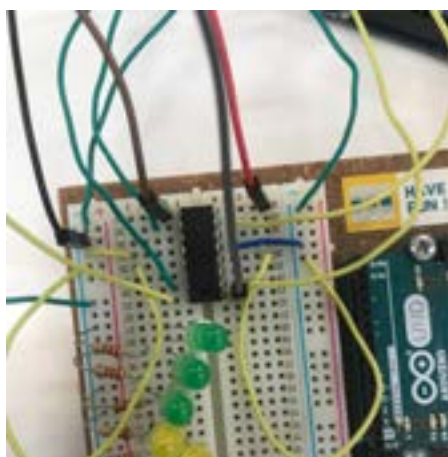
Create a sample circuit for one chip on a bread board. (Use a bread board so the chips are easily changeable). Then power them with sample program from Arduino. And repeat to check all the chips.

Results

All the chips works well except for one. But the issue the chip had was that it required power to an extra pin as well as the others. This was a weird anomaly that was lucky to find.

Conclusion

All chips will be usable in the project.



Test 2

Aim

Make sure all ports on the Arduino can supply power after having the main female adapter pins removed.

Method

Tell the Arduino to output power from all the pins which require testing. Then connect one contact of a multimeter to ground then place the other contact on each pin.

Results

All pins were usable some had to still have some old solder removed.

Conclusion

All pins will work without worry



Test 3

Aim

Make solid connections using solder to wire up the chips to the Arduino and make sure there is a common colour scheme for the wires.

Method

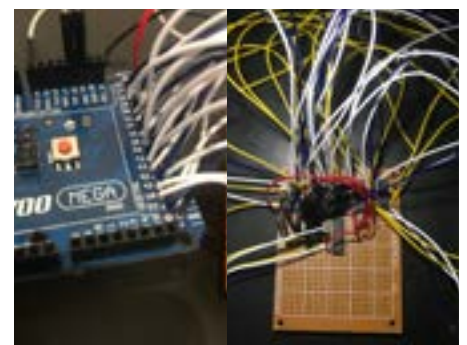
Start soldering and when an unforeseen problem occurs, solve it.

Results

During the wiring process, all the wires were soldered into their correct pins with all colours corresponding to the information that would be transmitted by the Arduino. But when tested with multimeter not all connections had were solid. This was fixed by going over with more solder until every connection was supplying electricity.

Conclusion

Wiring ready for next step.



Continued - Test 4

Aim

The requirements of this device are such that the suit needs to be able to be unplugged from the electronics box. So a custom plug port is required.

Method

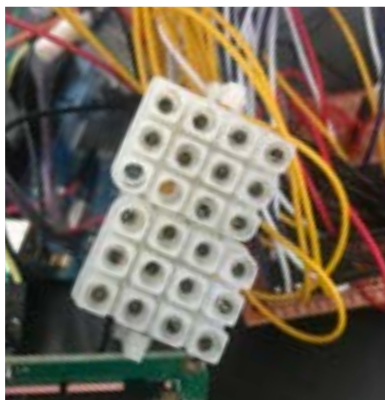
Solder each wire to an insertable connection for two 12 pin plugs. There is only the need for 22 connections though. Then test that these connections are solid.

Results

These connections are solid because there was zero resistance shown on the multimeter. But when the Arduino was used to control the power distribution no power was traveling through the pins. After research about all the components that may have issues. It turned out that L293D Chip had a minimum voltage requirement of 4.5V that needed to be supplied by an external power supply. And at this point in time the power that was being supplied was 5.2V from old AA batteries. So when more than two Chips need to run there would be a voltage drop to under the 4.5V required. The solution was to just have a power supply supplying more voltage.

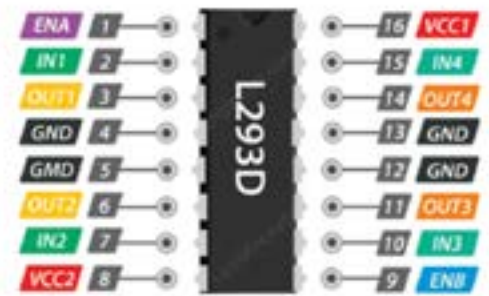
Conclusion

All the components in the circuit for the haptic pods are working well and ready to be put to use.

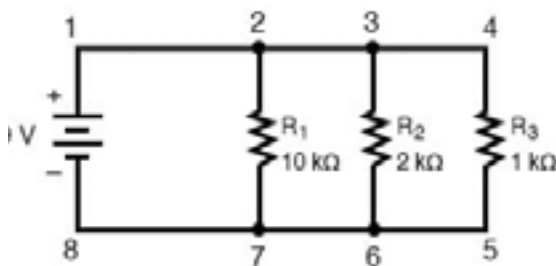


L293D Motor Driver

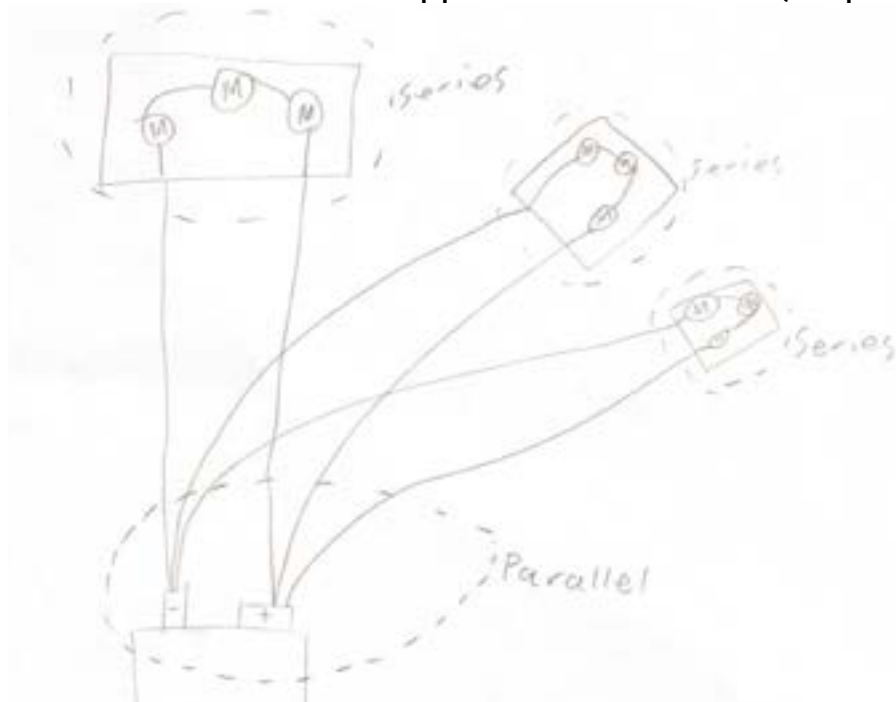
The L293D motor driver chip is usually a chip that is bought already inside a premade circuit board. But these boards are half the length and same height and width as an Arduino Mega which makes them reasonably large. This is usually ok for a hobbyist as they only need maximum 1 as that drives 2 motors. But this project needed 6 to run 11 haptic pods. Having 6 driving boards is not economical with space inside the case. So the only solution to this is to make 1 control board out of 6 driving chips. In pin VCC1 the chip requires 5V and in VCC2 the chip requires 4.5V to 36V



Circuits in parallel

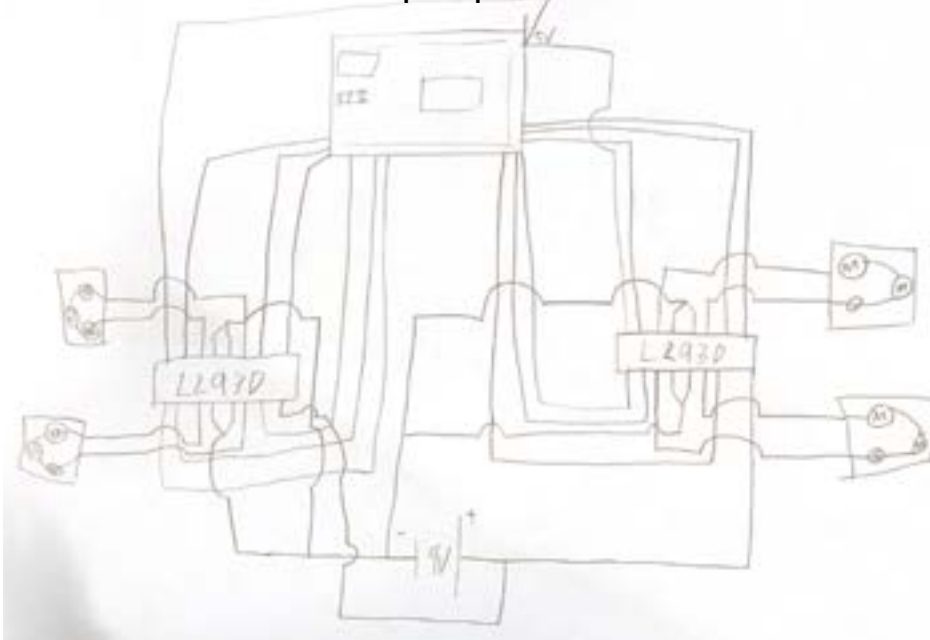
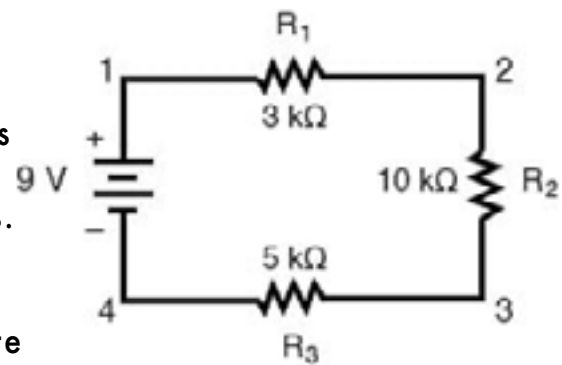


The reason why these chips are required is because in each haptic pod there are 3 small haptic motors which require 1.5V to 3V, and if supplied with less than that they will not work. The way to calculate how many volts are required to run a haptic pod depends on one main factor. Are the motors in parallel or series? When components are in parallel their positive and negative terminals are in contact. This makes the voltage required unchanged to the initial which in this case is 1.5V but the current required is timesed by the number of components. This is not a viable way to run motors that burn out when the voltage exceeds 3V as the Arduino supplies 5V from all its I/O pins.



Circuits is series

So, the haptic pod needs to be wired in series. When components positive terminal is connected to power and the next components positive is connected to the previous ones ground. When components are in series the current required stays unchanged but the voltage required is timesed by the number of components. This is a viable way to wire the haptic pod because it means each pod draws 1.5V to $3V \times 3 = 4.5V$ to 9V. This is good because most electronics can operate within this region. But as there are 11 haptic pods if connected directly to an Arduino, this has the potential to short out the Arduino if for example a motor gets stuck and draws too much current. So it is best to make a new circuit for each haptic pod so they can draw as much current as needed. This is done by using the L293D chip and wiring up an external power supply which means the Arduino only needs to tell the chip what actions the motor should be doing. It also means the motors can be run at a higher voltage to allow for more vibration of the haptic pods.



Haptic pods wired to battery

The haptic pods are in parallel to each other so they can be controlled separately and it does not matter how much current they draw because it comes directly from a battery which can theoretically discharge almost instantly supplying any amount of current required. But batteries are design to have Internal Resistance which just makes the battery safer to operate this also inhibits its ability to discharge in an instant. But that is not an issue as this project is not drawing that much current.

Testing and development of lidar

Camera and Raspberry Pi Testing

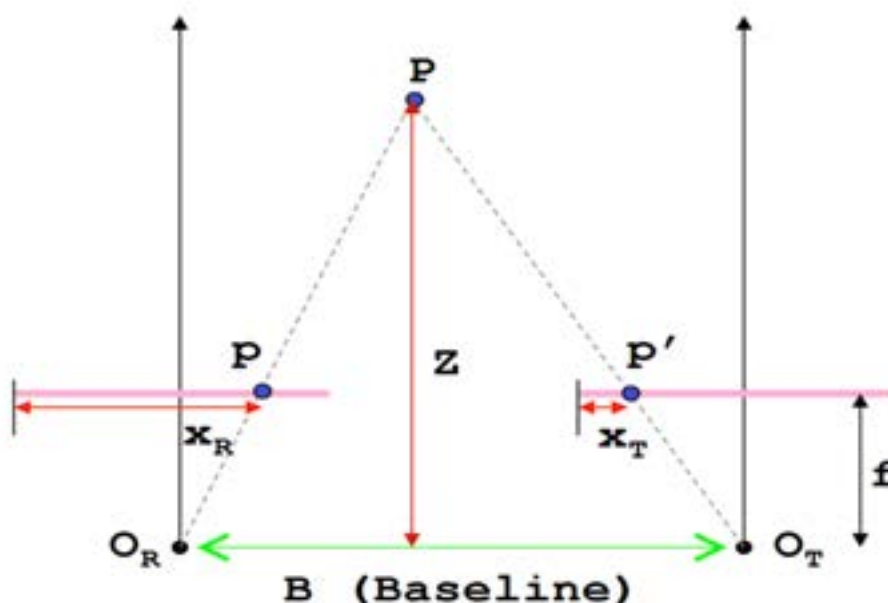
Originally this project was looking at using cameras to gauge the distance of cars and objects from the rider. To do this a Raspberry Pi was going to be used. A program that was going to be used, was written in python using OpenCV. OpenCV is a way to be able to identify what an object is by using image recognition and then determine the distance of that object by using triangulation made possible by having two cameras mounted x distance apart. The Raspberry Pi was the micro controller of choice because it is basically a tiny computer that runs a Linux OS designed specifically for these applications called Raspbian.

Aim

To see if a Raspberry Pi with two cameras can determine a specified objects distance and can control DC motors with this information

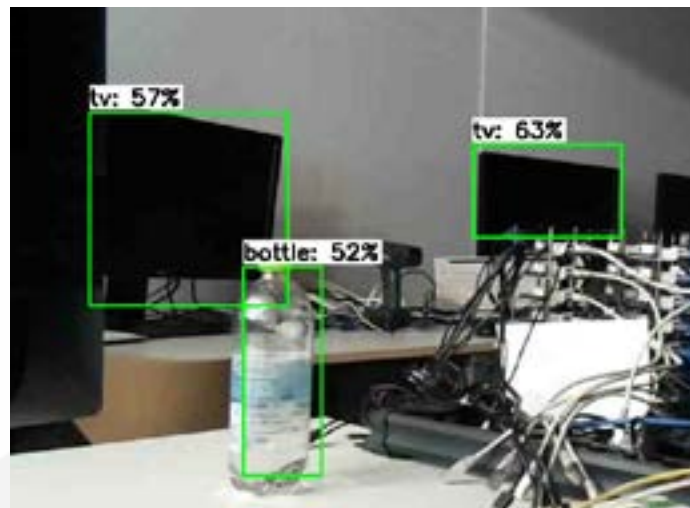
Method

Write a program on python using public repositories to help identify an objects nature this will make the Raspberry Pi recognise the specified objects. Then determine their distances by using triangulation which is done by comparing how many pixels difference and object appears in each camera then use trigonometry with the known distance B between the two cameras to determine z (which is the distance). With this information implement an action.



Results

The Raspberry Pi had no issues operating the DC motors and it was also able to run the image recognition program written but as it was running it was incredibly slow and would not be fast enough at all for a safety situation. This may have been able to be fixed if for example a machine learning program was written, and the Raspberry Pi could create short cuts withing the program therefore being capable of quick recognition. When the second camera was used the Raspberry Pi struggled so much so that within a couple seconds just running the recognition program froze until quit.



Conclusion

The **Raspberry Pi is not a suitable processing device** for projects which include image processing. There may be a way to write an incredibly light version of these programs which is possible as they are used in phones. But after the distance program and implementation programs were written into final program the Raspberry Pi does not seem likely to hold up under that stress. This leaves the project needing another way to calculate distance.



Testing ultrasonic Ranger

Ultrasonic Rangers are a type of device that determines the distance of open air directly in front of them. Ultrasonic devices use sound to transmit a signal, called the chirp, which will reflect off a surface, called the echo, and then be picked up again by a receiver on the device. The time that the signal took to travel back to the ranger is calculated as $d(\text{distance}) = v(\text{velocity of sound in air}(331 + (0.6 \times T \text{ Degrees Celsius})) \times t(\text{time}))$.

Aim

To see if it is suitable to use an ultrasonic sensor to measure distance accurately and implement actions in program to identify this.

Method

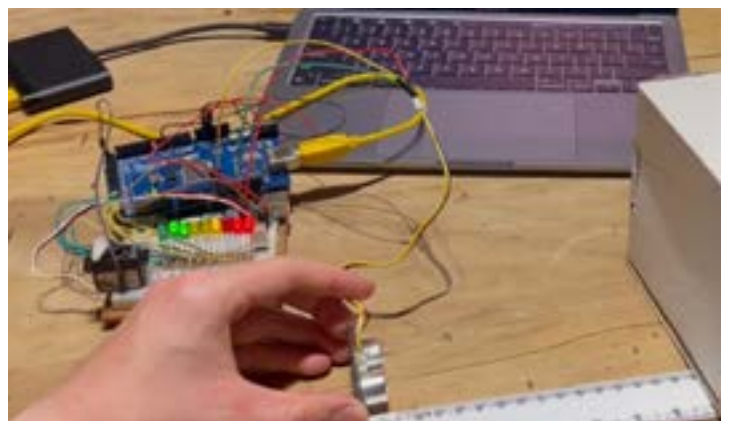
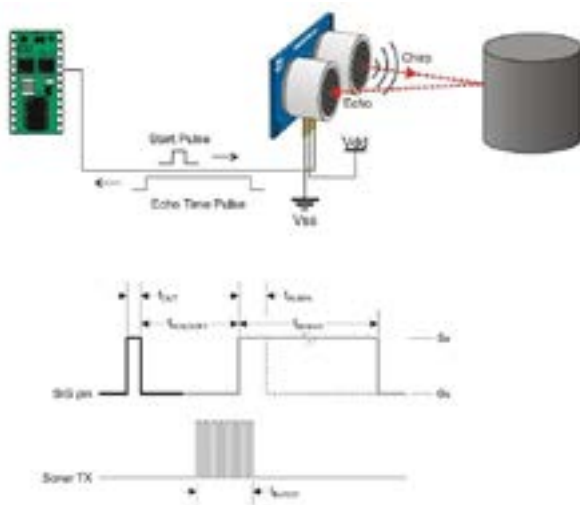
Use the ultrasonic library which is in the Arduino IDE folders. Then write an if variable to turn on a series of lights as distance changes.

Results

The ultrasonic ranger managed everything required. But its range is only 6 meters.

Conclusion

The ultrasonic ranger is very reliable but it does not measure accurately after 6 meters. This is still a very usable component. This ranger can be used as a blind spot detector because when something is in the blind spot they will always be within 6 meters of the rider.



Testing Lidar

Lidar(Light Detection And Ranging) is a way to determine the distance of a surface in front of the ranger. This type of ranging has a large application in self driving technologies and military technologies already. Lidar is a good way to range because the device is obtaining information at the speed of light. The actual technology works in a similar way to ultrasonic rangers. This device will emit a specific near-infrared laser beam which will be reflected off a surface and then received back at the device by a light receiver. This device is capable of many other applications such as measuring temperature and light absorption of a surface.

Aim

To measure and accurate distance which is far further than the ultrasonic ranger and be able to implement actions when at different distances.

Method

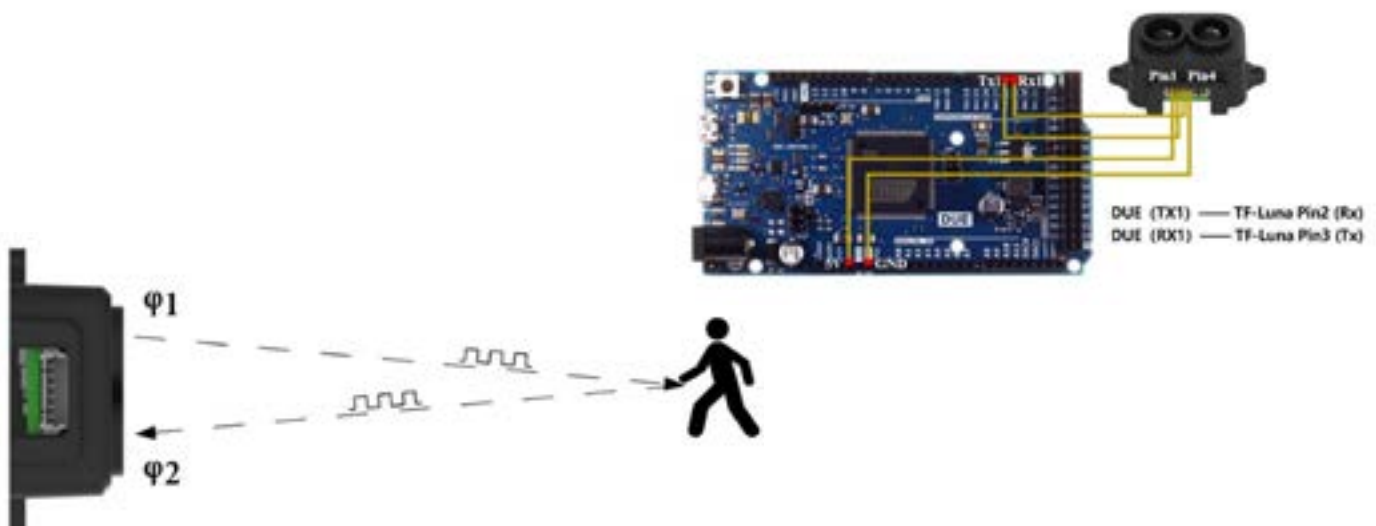
The lidar does not come with its own library to help people program it easily. So the program has to declare what the signals mean and how to display them so they are able to be interpreted by people and the rest of the code. Then measure to the half centimetre to make sure the lidar is accurate.

Results

Ma Eb j k p o n q e h f o n k b g m h / f m k l m k m e r b g j h h k l g j e h m doors. It was incredibly difficult to program and even find the correct pins of the TF Luna Lidar. But once it was programmed it was easy to great actions for constantly varying distances.

Conclusion

This is a great improvement to the range of the ultrasonic rangers and it would be good to use because of the large range and it will not be as effected by ambient noise in the form of near-infrared waves as the ultrasonic ranger can become by other ultrasonic waves.



Testing of suit development

Aim

To find a way to sew neoprene well and neatly.

Method

Try multiple ways of sewing using different sewing settings and different needles. Also ask professional sewers.

Results

A walking foot turned out to be a good way to make good stitches each oscillation. The use of a titanium embroidery needle also help drastically.

Conclusion

Sewing was a major part of this project and proved to be quite a challenge when it came to sewing more than one layer of neoprene. Neoprene is a spongy rubber compound covered in a synthetic layer on the front and back side. Neoprene is stretchy and because of this stretch the neoprene often would hold the needle not allowing it to penetrate the bottom layer. To combat this the use of a walking foot that fitted the machine I was using was suggested by Sharon Hurry. The walking foot moved the fabric instead of a conventional foot which drags and pulls the fabric. This specific fabric movement allowed for the sharpest titanium embroidery needle to penetrate fully each oscillatory motion.



The thread would not catch with the standard stationary foot. This lack of catching made sewing with neoprene almost impossible.



This walking foot was able allow the thread to catch because of its ability to move the fabric evenly when sewing. This foot works so well because it has teeth on both top and bottom which move independently.

Bias Binding

Because neoprene is a very hard material to cut and sew evenly there needs to be something to cover the edges where the cuts are most visible.

Aim

Improve the look of the edges of the neoprene.

Method

Try overlocking and sewing bias binding made from elastic fabric to the edge of three layers of neoprene.

Results

Overlocking neoprene was alright when it was 2 layers but it sometimes did not catch, but when it got to 3 layers of neoprene the overlocker could not handle a fabric that tough. Using bias binding was the other option, to make the bias binding I used the elastic fabric that was also used in the Haptic Pod pockets. To make the bias binding I evenly cut long strips of the elastic fabric and the ironed creases into it. Then sewed the fabric trying to show as little thread as possible.

Conclusion

The bias binding was better than overlocking but will take longer to do and will be more difficult to do. But the result will be much better in the end.



Neoprene Testing

Cutting neoprene is very easy to mess up and finding the correct tool is essential. In my tests for what I could use to cut neoprene I brainstormed all the potential tools and techniques. It is essential for this project that the neoprene has neat edges.

Tools that can be used :

- Box cutter
- Scalpel
- Scissors
- Laser Cutter

Aim

Find tool that can cut neoprene neatly and precisely

Method

Brainstorm consequences then cut the neoprene

Box Cutter



Sharp Scissors



Scalpel



Results

- 1) The Box cutter has no consequences to using it but it pulled and frayed the outer fabric
- 2) The scalpel did not fray the fabric as much but still frayed it
- 3) Scissors had potential of contorting the fabric before cut which some pairs did but when testing finished the sharpest fabric scissors were able to hold up best
- 4) The laser cutter did not make it past the brain storm phase because even though it would make the neatest cuts. This is because after research I discovered that when neoprene is burned it can produce hydrogen cyanide gas, mustard gas and other slightly less harmful fumes. These fumes are incredibly harmful usually used as Weapons Of War. But a laser cutter would vaporises the neoprene which may mean it does not release any harmful fumes. But for safety the neoprene was not laser cut just in case.



Conclusion

Sharp Scissors came out on top when testing because they cut neoprene with the neatest slice without causing a danger.

Haptic Pod pocket

The primary purpose of this suit is to press haptic pods against the user's body. These pockets need to be able to hold the pods steady while not dampening any of vibrations as this is the essential form of communication between sensors and user.

Aim

Make a pocket that will fit haptic pods securely and not dampen vibrations. Make it so these pods are removable so the suit can be washed. Brainstorming ways in which this can be done. Then make sectioned models to see if it will work.

Method

The best way to make a pocket that will apply pressure to the back is to use a stretchy solid material such as neoprene and use two layers. Then sandwich two layers one which has a hole cut out that is the size of the Haptic pod with a synthetic elastic protruding from inside the hole. When the haptic pod is inside this will make it sit against the body without being uncomfortable for the user. But the pocket still needs to be able to be removed and to do this, on the other layer there needs to be a slit cut into it that is just under the width of the pod so it cannot fall out.

Results

This design was a great success it was able to house the haptic pod easily and it did not move around and it applied the perfect amount of pressure when testing. This pressure requirement may be different when in the actual suit, but the suit should have the ability to change the pressure. The pocket does not dampen any of the vibrations, because by supporting the haptic pod the pod does not move freely which focuses the vibrations to the body.

Conclusion

This Haptic Pod pocket will function very well when in the suit.



Cable Management

As this suit has electronics all through it, it requires cable management, so cables are not on the outside and getting caught on things.

Aim

To hide cables and make them resistant to pulling. Keep them accessible.

Method

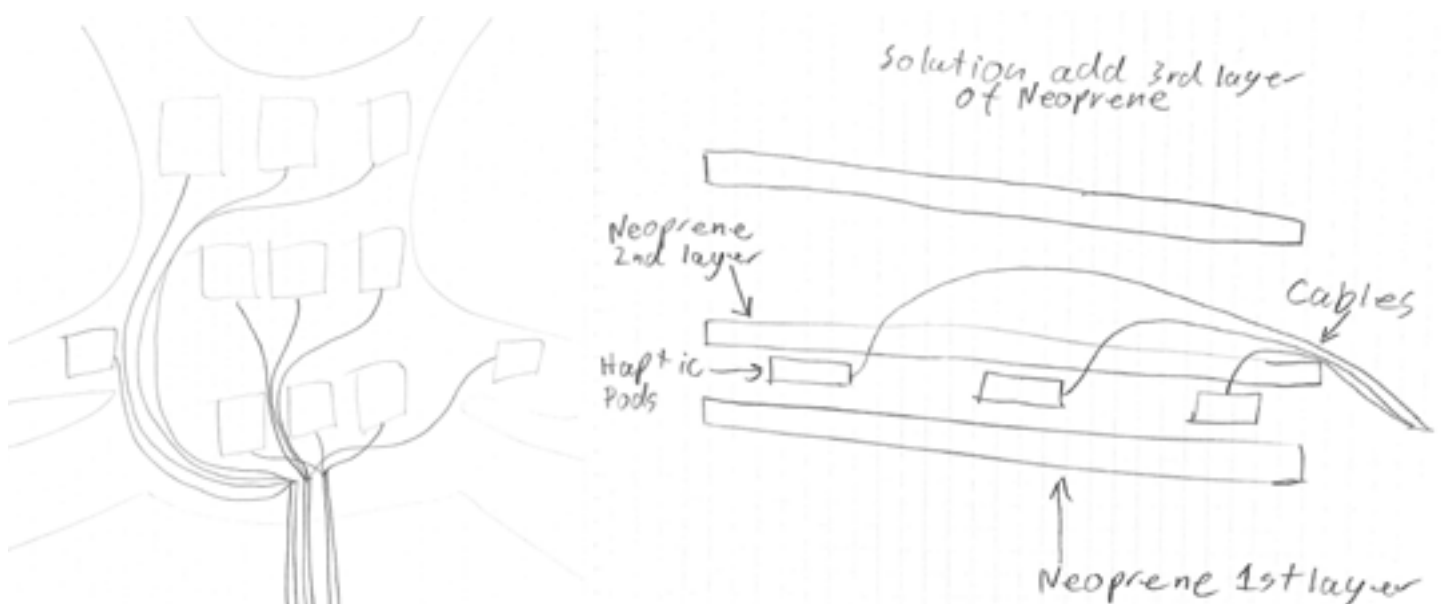
Find a way to keep the cables safe during operation and make the solution aesthetically pleasing. Consider many ways this can be done.

Results

To hide the cables and make them pull resistant while keeping them accessible came down to adding a third layer to the suit which would be a blank layer that would just cover the previous layer which has the slits in it. The previous layer was the layer that had the cords protruding out of it because it is the layer that holds in the haptic pods. The final layer is to go on the previous layer and direct the cables down to the base of the suit to then join as one group of cables. To make the cables resistant to pulling, at the congregation of the wires at the base of the suit there needs to be a Velcro piece which holds them in place.

Conclusion

There shall be a third layer which covers, directs and holds the cables.



Application of conclusions

Haptic Pod development

After testing variations of different haptic pod ideas the use of one that is smaller but with 3 motors, all at 45 degrees from each other, has the best effect when transmitting information through vibrations to the user.

Fabric For Suit

To make a suit that applies pressure on the haptic motors but is still comfortable to wear, I have concluded that neoprene will be the fabric that is used for this application because it is a solid fabric but is still able to stretch and is not affected by moisture. I will also use Sports mesh polyester because it also stays unaffected by moisture and has multidirectional stretch.

Design of suit

To get a good design for the suit. I will use a card board mock-up that has been shaped to fit me, but also been designed to be adjustable and have a universal shape. This mock-up has good placement for the haptic pods.

Straps for suit

For the suit I will have straps extruded from the bottom two rows of haptic pods so that they can have adjustable pressure on them. The straps will be on an angle down to wrap around the body neatly.

Fender box

After gathering information on the shapes of motorcycle fenders I have come to a decision on the shape of the fender box. The fender box will be a rectangular rear fender with a curvature of 35mm below the apex. This box is to be able to fit all the electronics that the project entails. Taking inspiration from aerodynamic design the fender box will fit all the requirements set.

Haptic motor

The haptic motor chosen for this project will be the ERM Motors as they create vibrations in two directions compared to an LRA which only makes vibrations in one direction.

Electronics tools

The tools that will be used to develop the electronics will be most importantly a soldering iron. And then a sparrow wire stripper which will make stripping wires a simple job compared to how it often is very difficult.

Program

This project will use code from multiple different tests that have occurred. This code will be a congregation of many other pieces of code mashed into one program.

Wiring

The wiring of the electronics is going to be as seen in the diagram because this is what will work with all the various components.

Haptic Pod wiring

The haptic pods will be wired up in series with each other using six L293D motor controller chips which will have power supplied to it from an external source in parallel. The power will be supplied from an external source from the Arduino because this many components could potentially draw too much current from the Arduino and kill the project.

Haptic pod connecting port

The haptic pod connecting port is needed because if the haptic pods need to be able to be detached from the rest of the electronics. Two 12 pin ports will be used which and this project will use 22 ports.

Lidar

After research and testing using a camera and raspberry pi to do image recognition and discovering that the raspberry pi cannot handle that much processing, I have made the conclusion to use an Arduino, lidar and ultrasonic ranger to measure distances.

Sewing neoprene

To sew neoprene after testing I discovered that the only way to sew adequately with a normal sewing machine is to use a walking foot and titanium embroidery needle. This is the only way that the machine can catch the fabric.

Edges of neoprene

I discovered that neoprene after being sewn ends up with bad looking edges. So to fix this I discovered that using bias binding cut from polyester sports mesh works at covering the edge impurities.

Cutting neoprene

To cut neoprene neatly requires a very sharp pair of scissors.

Haptic pod pocket

To make a haptic pod pocket that is to be in the suit design it will be two layers of neoprene, one with a hole cut in the front and polyester sports mesh protruding from it, and then the other side will be the other piece of neoprene with slices cut into it to allow the haptic pod to be inserted and removed.

Cable management

To manage the cables which come from the second layer of neoprene one more layer is to be put on top to direct the cables the base of the suit. Where a piece of Velcro will hold all the cables in place.

Identification and Justification of ideas and resources used

Haptic pods

Although it is not a haptic pod, haptics are also used largely in the gaming world in controllers and now the new gaming suits. The reason why they are used by others and myself is because it allows a computer to transfer information through to a user without using sound or sight.

Sensor box and Arduino and integrated components

Sensor boxes may be a car, a phone, a Geiger counter or in this case rear view and blind spot detection and ranging. There would not be any other way to create an electronic project of this nature without using electronics and sensors to identify the information you want to relay.

Soldering

Soldering is a great way to make solid connections between electronic components

Sewing tools

Using a sewing machine is great technology that can increase the speed of production by a lot. But when there needs to be a neat line of stitching through something the plain needle and thread will do the job very well because of the control that is gained by this technique.

Textile materials

To make this project into a project that could be demonstrated after the electronics are finished, I needed to make a suit that was comfortable and fitted the requirement of assisting the relay of information through haptic pods. A good material would probably be one which is synthetic and unaffected by moisture.

3D Printing

3D printing is one of the most useful prototyping resources to have when developing various devices. This project would not be able to be completed well without the use of 3D printers.

Human Recourses

I was very lucky to be able to meet with Sharron Hurry (A professional sewer) to try and use her industrial sewing machine. But unfortunately, her industrial sewing machine only was able to sew well through one layer of neoprene. But Sharron did suggest using a walking foot on a machine I had at home. This suggestion saved the outcome of this project. Without using a walking foot I would have either had to hand stitch the whole suit or have to put up with faulty stitches.

Tools used to develop electronics

To make all the electronics, I used leadless solder to make solid connections between all the conductive material. The soldering iron that was used was a HAKKO soldering iron. This iron is incredibly good because it heats up to 350 degrees Celsius in under 20 seconds and maintains this heat for as long as desired.



An incredibly underrated tool that was used for the creation of this device, is the sparrow wire stripper and crimper. This tool has proven to be endlessly useful. It makes the task of stripping wires exponentially easier and quicker. As a significant amount of this project is the manipulation of insulated wires.



Evidence and application of practical skills to produce a quality project

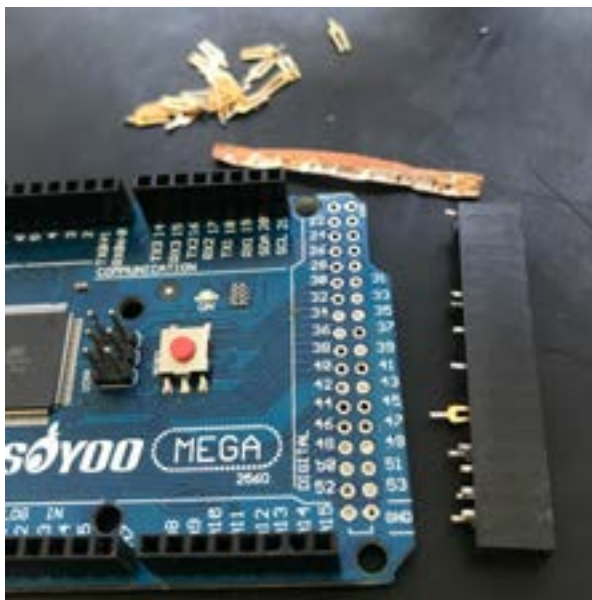
Development of Electronics

Gathering of electronic components



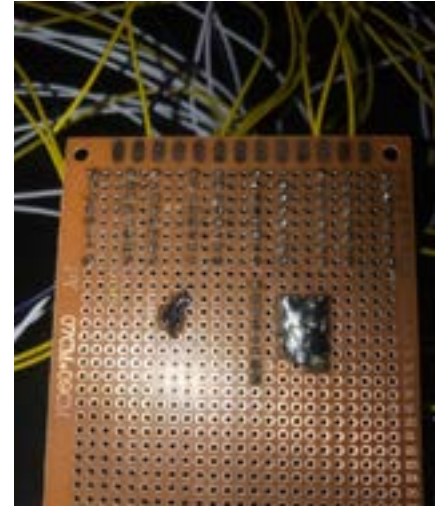
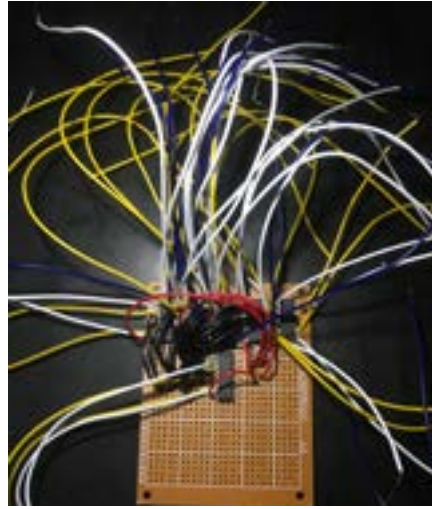
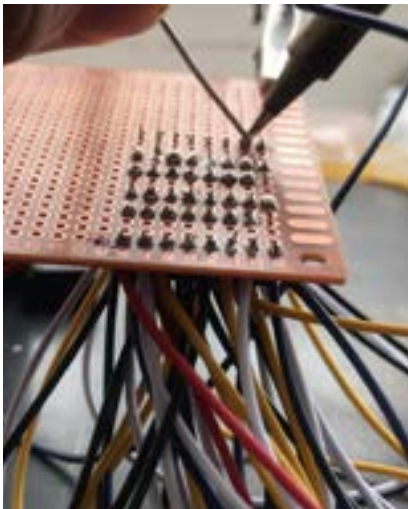
To desolder The Pins which I wanted to make a permanent connection to that had already been soldered by a robot. I had to use desoldering wick which acts as a heat sink and the solder seeps into it.

Prepare Arduino for permanent wiring by desoldering



Desoldering wick

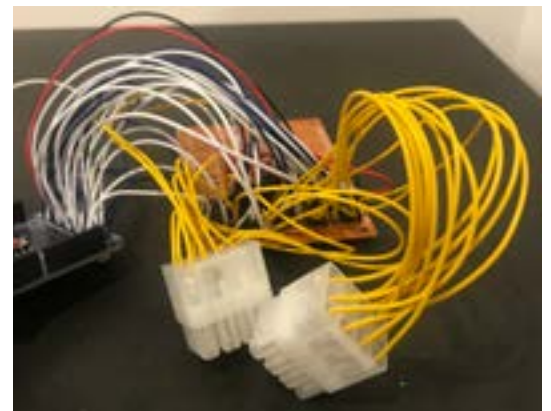
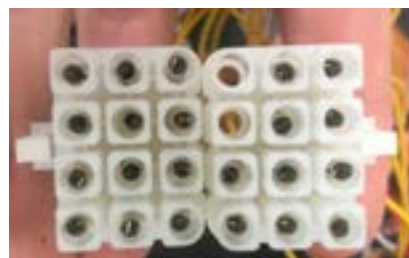
Soldering L293D Driving chip to solderable bread board



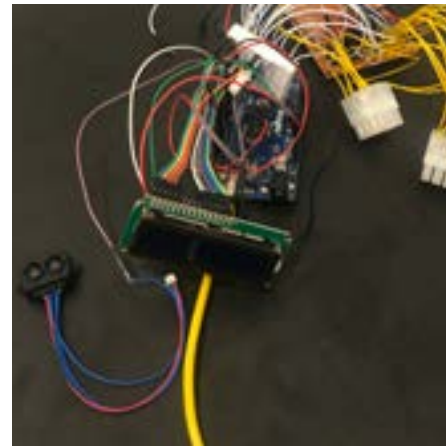
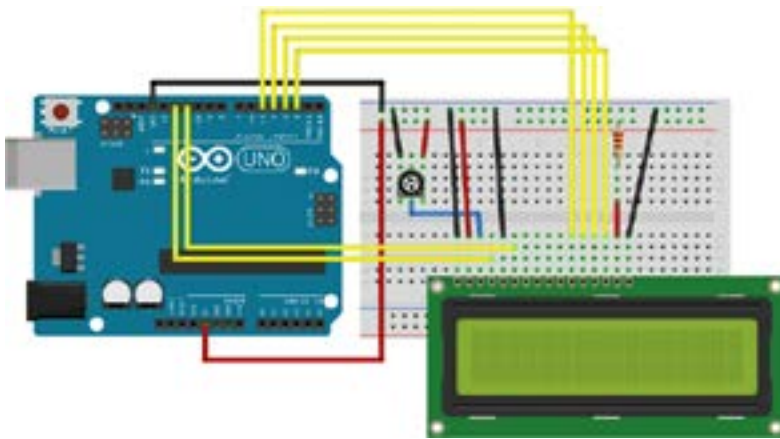
Solder new wires into removed pins in the wiring scheme determined from the planning phase



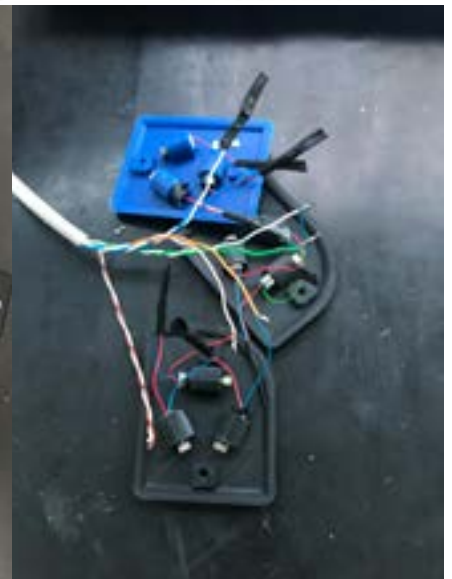
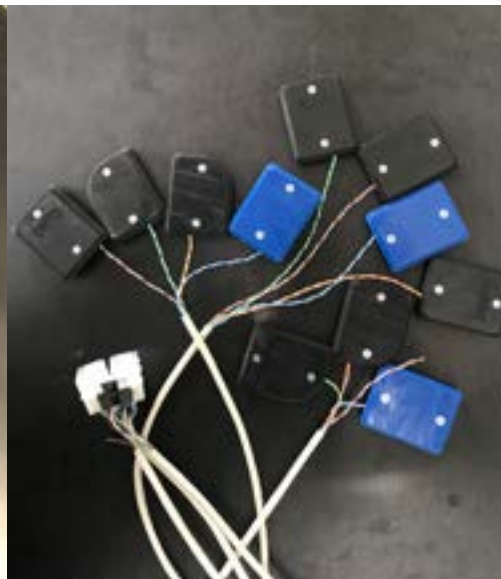
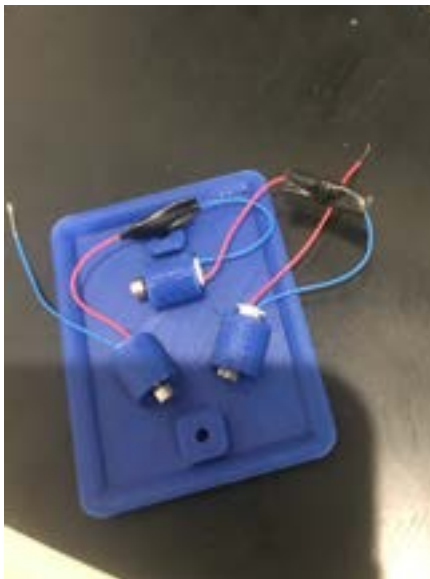
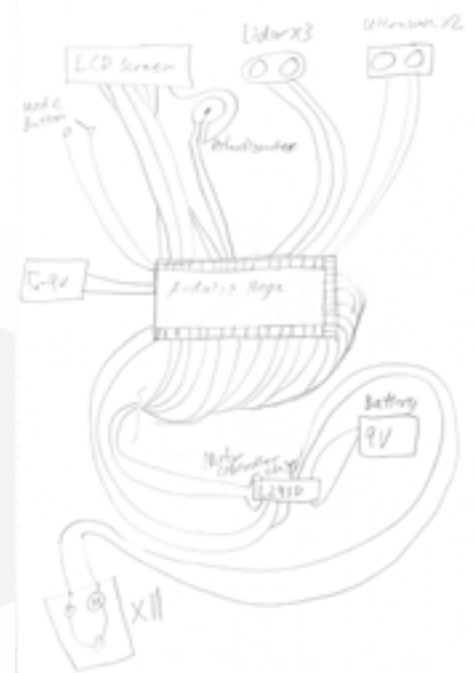
Connecting Motor driving chips to Detachable port for ease of use to easily connect and disconnect Haptic Pods



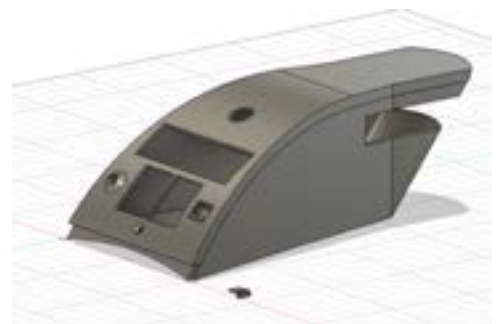
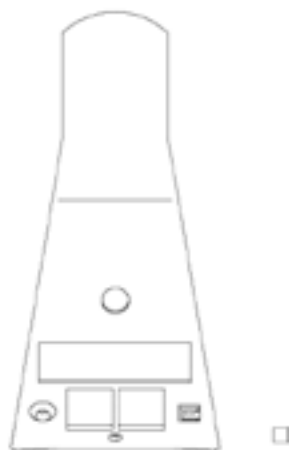
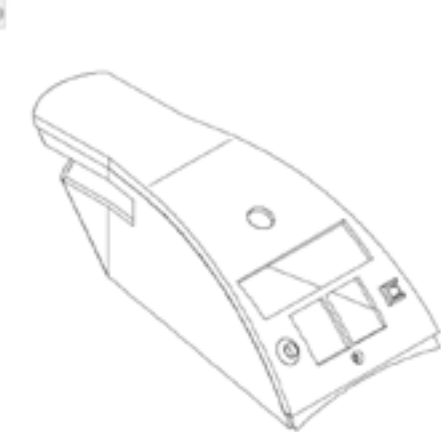
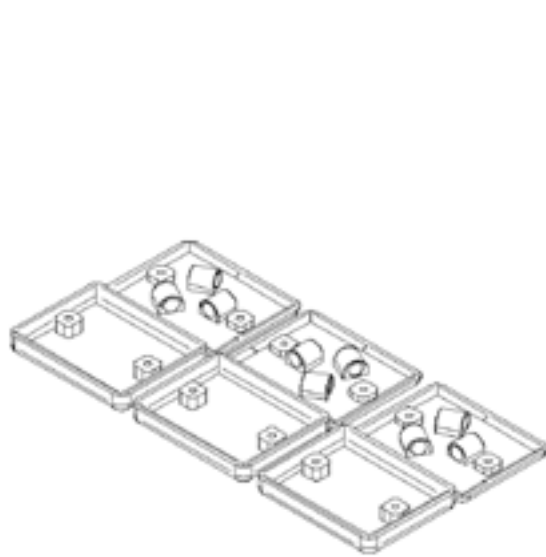
Attach Lcd Screen and sensors



Making haptic motors with corresponding plug that is on the areduino



Designing Haptic Pods and Fendor Box on Fusion 360 to then be 3D printed



```
MotoTest | Arduino 1.8.14 Hourly Build 2021/04/09 02:34

Final Code made from
all the tests

MotoTest §
#include <SoftwareSerial.h>
#define SoftwareSerial Serial1(18,19);
#include <Servo.h>
#include <LiquidCrystal.h>

const int rs = 12, en = 11, d4 = 5, d5 = 4, d6 = 3, d7 = 2;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);

Servo servo;

int dist;
int check;
int i;
int uart[9];
const int HEADER=0x59;

//Mid Motors
int enMidBotMot = 22;
int MidBotMot1 = 23;
int MidBotMot2 = 24;

int enMidMidMot = 25;
int MidMidMot1 = 26;
int MidMidMot2 = 27;

int enMidHigMot = 28;
int MidHigMot1 = 29;
int MidHigMot2 = 30;

//Left Motors
int enLefBotMot = 31;
int LefBotMot1 = 32;
int LefBotMot2 = 33;

int enLefMidMot = 34;
int LefMidMot1 = 35;
int LefMidMot2 = 36;

int enLefHigMot = 37;
int LefHigMot1 = 38;
int LefHigMot2 = 39;

//Right Motors
int enRigBotMot = 40;
int RigBotMot1 = 41;
int RigBotMot2 = 42;

int enRigMidMot = 43;
int RigMidMot1 = 44;
int RigMidMot2 = 45;

int enRigHigMot = 46;
int RigHigMot1 = 47;
int RigHigMot2 = 48;

//Blind Spot Motors
int enBlirig = 49;
int Blirig1 = 50;
int Blirig2 = 51;

int enBlilef = 52;
int Blilef1 = 53;
int Blilef2 = 69;

//MID Section
analogWrite(enMidBotMot, 255);
digitalWrite(MidBotMot1, HIGH);
digitalWrite(MidBotMot2, LOW);
analogWrite(enMidMidMot, 255);
digitalWrite(MidMidMot1, HIGH);
digitalWrite(MidMidMot2, LOW);
analogWrite(enMidHigMot, 255);
digitalWrite(MidHigMot1, HIGH);
digitalWrite(MidHigMot2, LOW);

- LEFT Section
analogWrite(enLefBotMot, 255);
digitalWrite(LefBotMot1, HIGH);
digitalWrite(LefBotMot2, LOW);
analogWrite(enLefMidMot, 255);
digitalWrite(LefMidMot1, HIGH);
```

Final Code made from all the tests

Libraries that are included in the code to run physical components

Declaring the pins of the arduino that will run LCD

Creating Integers which is a type of data storage on Arduinos

Declaring which ports each Haptic pod is connected to.

Naming convention [COLUMN][ROW]

Needed access to the bare ports so lengths of wire could be soldered directly to the board without needing a male adapter. This makes production of Haptic pod conector far easier




```
digitalWrite(LefMidMot1, HIGH);
digitalWrite(LefMidMot2, LOW);
analogWrite(enLefMidMot, 255);
digitalWrite(LefHigMot1, HIGH);
digitalWrite(LefHigMot2, LOW);
```

```
- RIGHT Section
analogWrite(enRigBotMot, 255);
digitalWrite(RigBotMot1, HIGH);
digitalWrite(RigBotMot2, LOW);
analogWrite(enRigMidMot, 255);
digitalWrite(RigMidMot1, HIGH);
digitalWrite(RigMidMot2, LOW);
analogWrite(enRigHigMot, 255);
digitalWrite(RigHigMot1, HIGH);
digitalWrite(RigHigMot2, LOW);
```

```
- Left BLIND Spot
analogWrite(enBliLef, 255);
digitalWrite(BliLef1, HIGH);
digitalWrite(BliLef2, LOW);
```

```
- Right BLIND Spot
analogWrite(enBliRig, 255);
digitalWrite(BliRig1, HIGH);
digitalWrite(BliRig2, LOW);
*/
```

```
void setup()
```

```
{
  servo.attach(8);
  servo.write(90);
  lcd.begin(16, 2);
```

```
  Serial.begin(9600);
  Serial1.begin(115200);
```

```
  pinMode(enMidBotMot, OUTPUT);
  pinMode(MidBotMot1, OUTPUT);
  pinMode(MidBotMot2, OUTPUT);
```

```
  pinMode(enMidHigMot, OUTPUT);
  pinMode(MidHigMot1, OUTPUT);
  pinMode(MidHigMot2, OUTPUT);
```

```
  pinMode(enLefBotMot, OUTPUT);
  pinMode(LefBotMot1, OUTPUT);
  pinMode(LefBotMot2, OUTPUT);
```

```
  pinMode(enLefMidMot, OUTPUT);
  pinMode(LefMidMot1, OUTPUT);
  pinMode(LefMidMot2, OUTPUT);
```

```
  pinMode(enLefHigMot, OUTPUT);
  pinMode(LefHigMot1, OUTPUT);
  pinMode(LefHigMot2, OUTPUT);
```

```
  pinMode(enRigBotMot, OUTPUT);
  pinMode(RigBotMot1, OUTPUT);
  pinMode(RigBotMot2, OUTPUT);
```

```
  pinMode(enRigMidMot, OUTPUT);
  pinMode(RigMidMot1, OUTPUT);
  pinMode(RigMidMot2, OUTPUT);
```

```
  pinMode(enRigHigMot, OUTPUT);
  pinMode(RigHigMot1, OUTPUT);
  pinMode(RigHigMot2, OUTPUT);
```

```
  pinMode(enBliRig, OUTPUT);
  pinMode(BliRig1, OUTPUT);
  pinMode(BliRig2, OUTPUT);
```

```
  pinMode(enBliLef, OUTPUT);
  pinMode(BliLef1, OUTPUT);
  pinMode(BliLef2, OUTPUT);
```

Demonstration of code for any person who may want to re-code some actions. This text does not do anything to program as it is blank text. If a section got copied into the looped code it would operate the corresponding haptic pods.

This void (congregation of code) prepares the arduino and attached devices for operation.

Sets the servo and starts the LCD Screen.

Starts two communication channels between computer and arduino.

This is telling each port if they are giving or receiving 5 Volts.

```

lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Feeling sight");

lcd.setCursor(0, 1);
lcd.print("MDP Testing");

```

printing initial text on the LCD screen.

```

}

```

Void will loop until device is reset or disconnected from power.

```

void loop()
{

```

```

    long Dcm;
    dist = uart[2] + uart[3] * 256;
    Serial.print("    ");
    Serial.print(dist, DEC);
    servo.write(90);{

```

Tells the arduino what distance will look like from the information that it will be given from the lidar. Then it tells the arduino to send this information to the computer if there is one.

```

        analogWrite(enMidBotMot, 255);
        analogWrite(enMidMidMot, 255);
        analogWrite(enMidHigMot, 255);

```

Starts the enable pins that run the L293D Motor Driver chip

```

        if((Dcm < 200 && Dcm > 100)){
            digitalWrite(MidBotMot1, HIGH);
            digitalWrite(MidBotMot1, LOW);
            digitalWrite(MidMidMot1, LOW);
            digitalWrite(MidMidMot2, LOW);
            digitalWrite(MidHigMot1, LOW);
            digitalWrite(MidHigMot2, LOW);
        }

```

Asks the arduino if the distance is between the given range.

```

        else if((Dcm <= 100 && Dcm > 50)){
            digitalWrite(MidBotMot1, HIGH);
            digitalWrite(MidBotMot1, LOW);
            digitalWrite(MidMidMot1, HIGH);
            digitalWrite(MidMidMot2, LOW);
            digitalWrite(MidHigMot1, LOW);
            digitalWrite(MidHigMot2, LOW);
        }

```

if answer is yes, run program, if not go to next variable.

```

        else if(Dcm <= 50){
            digitalWrite(MidBotMot1, HIGH);
            digitalWrite(MidBotMot1, LOW);
            digitalWrite(MidMidMot1, HIGH);
            digitalWrite(MidMidMot2, LOW);
            digitalWrite(MidHigMot1, HIGH);
            digitalWrite(MidHigMot2, LOW);
        }

```

```

        else{
            digitalWrite(MidBotMot1, LOW);
            digitalWrite(MidBotMot1, LOW);
            digitalWrite(MidMidMot1, LOW);
            digitalWrite(MidMidMot2, LOW);
            digitalWrite(MidHigMot1, LOW);
            digitalWrite(MidHigMot2, LOW);
        }

```

```

        delay(250);
    }

```

```

    dist = uart[2] + uart[3] * 256;
    Serial.print("    ");
    Serial.print(dist, DEC);
    servo.write(135);{

```

Repeats the same distance detection and motor control as above but controls the left column motors.

```

        analogWrite(enLefBotMot, 255);
        analogWrite(enLefMidMot, 255);
        analogWrite(enLefHigMot, 255);

```

```

        if((Dcm < 200 && Dcm > 100)){
            digitalWrite(LefBotMot1, HIGH);
            digitalWrite(LefBotMot2, LOW);
            digitalWrite(LefMidMot1, LOW);
            digitalWrite(LefMidMot2, LOW);
            digitalWrite(LefHigMot1, LOW);
            digitalWrite(LefHigMot2, LOW);
        }

```

```

        else if((Dcm <= 100 && Dcm > 50)){
            digitalWrite(LefBotMot1, HIGH);
            digitalWrite(LefBotMot2, LOW);
            digitalWrite(LefMidMot1, HIGH);
            digitalWrite(LefMidMot2, LOW);
            digitalWrite(LefHigMot1, LOW);
            digitalWrite(LefHigMot2, LOW);
        }

```

```

    }
}

```



```

else if(Dcm <= 50){
  digitalWrite(LefBotMot1, HIGH);
  digitalWrite(LefBotMot2, LOW);
  digitalWrite(LefMidMot1, HIGH);
  digitalWrite(LefMidMot2, LOW);
  digitalWrite(LefHigMot1, HIGH);
  digitalWrite(LefHigMot2, LOW);
}
else{
  digitalWrite(LefBotMot1, LOW);
  digitalWrite(LefBotMot2, LOW);
  digitalWrite(LefMidMot1, LOW);
  digitalWrite(LefMidMot2, LOW);
  digitalWrite(LefHigMot1, LOW);
  digitalWrite(LefHigMot2, LOW);
}
delay(250);
}
dist = uart[2] + uart[3] * 256;
Serial.print(" ");
Serial.print(dist, DEC);
servo.write(180);{
  analogWrite(enBliLef, 255);
  if(Dcm < 200){
    digitalWrite(BliLef1, HIGH);
    digitalWrite(BliLef2, LOW);
  }
  else{
    digitalWrite(BliLef1, LOW);
    digitalWrite(BliLef2, LOW);
  }
  delay(250);
}
dist = uart[2] + uart[3] * 256;
Serial.print(" ");
Serial.print(dist, DEC);
servo.write(90);{
  analogWrite(enMidBotMot, 255);
  analogWrite(enMidMidMot, 255);
  analogWrite(enMidHigMot, 255);

```

Repeats the same distance detection and motor control as above but controls the Middle column motors.

```

if((Dcm < 200 && Dcm > 100)){
  digitalWrite(MidBotMot1, HIGH);
  digitalWrite(MidBotMot2, LOW);
  digitalWrite(MidMidMot1, LOW);
  digitalWrite(MidMidMot2, LOW);
  digitalWrite(MidHigMot1, LOW);
  digitalWrite(MidHigMot2, LOW);
}
else if((Dcm <= 100 && Dcm > 50)){
  digitalWrite(MidBotMot1, HIGH);
  digitalWrite(MidBotMot2, LOW);
  digitalWrite(MidMidMot1, HIGH);
  digitalWrite(MidMidMot2, LOW);
  digitalWrite(MidHigMot1, LOW);
  digitalWrite(MidHigMot2, LOW);
}
else if(Dcm <= 50){
  digitalWrite(MidBotMot1, HIGH);
  digitalWrite(MidBotMot2, LOW);
  digitalWrite(MidMidMot1, HIGH);
  digitalWrite(MidMidMot2, LOW);
  digitalWrite(MidHigMot1, HIGH);
  digitalWrite(MidHigMot2, LOW);
}
else{
  digitalWrite(MidBotMot1, LOW);
  digitalWrite(MidBotMot2, LOW);
  digitalWrite(MidMidMot1, LOW);
  digitalWrite(MidMidMot2, LOW);
  digitalWrite(MidHigMot1, LOW);
  digitalWrite(MidHigMot2, LOW);
}
delay(250);
}
dist = uart[2] + uart[3] * 256;
Serial.print(" ");
Serial.print(dist, DEC);
servo.write(45);{

```

```

analogWrite(enRigBotMot, 255);
analogWrite(enRigMidMot, 255);
analogWrite(enRigHigMot, 255);

if((Dcm < 200 && Dcm > 100)){
  digitalWrite(RigBotMot1, HIGH);
  digitalWrite(RigBotMot2, LOW);
  digitalWrite(RigMidMot1, LOW);
  digitalWrite(RigMidMot2, LOW);
  digitalWrite(RigHigMot1, LOW);
  digitalWrite(RigHigMot2, LOW);
}
else if((Dcm <= 100 && Dcm > 50)){
  digitalWrite(RigBotMot1, HIGH);
  digitalWrite(RigBotMot2, LOW);
  digitalWrite(RigMidMot1, HIGH);
  digitalWrite(RigMidMot2, LOW);
  digitalWrite(RigHigMot1, LOW);
  digitalWrite(RigHigMot2, LOW);
}
else if(Dcm <= 50){
  digitalWrite(RigBotMot1, HIGH);
  digitalWrite(RigBotMot2, LOW);
  digitalWrite(RigMidMot1, HIGH);
  digitalWrite(RigMidMot2, LOW);
  digitalWrite(RigHigMot1, HIGH);
  digitalWrite(RigHigMot2, LOW);
}
else{
  digitalWrite(RigBotMot1, LOW);
  digitalWrite(RigBotMot2, LOW);
  digitalWrite(RigMidMot1, LOW);
  digitalWrite(RigMidMot2, LOW);
  digitalWrite(RigHigMot1, LOW);
  digitalWrite(RigHigMot2, LOW);
}
delay(250);
}
dist = uart[2] + uart[3] * 256;
Serial.print(" ");
Serial.print(dist, DEC);
servo.write(180);{
  analogWrite(enBliRig, 255);
  if(Dcm < 200){
    digitalWrite(BliRig1, HIGH);
    digitalWrite(BliRig2, LOW);
  }
  else{
    digitalWrite(BliRig1, LOW);
    digitalWrite(BliRig2, LOW);
  }
  delay(250);
}
}

```

Repeats the same distance detection and motor control as above but controls the Right column motors.

Repeats the same distance detection and motor control as above but controls the Blindspot motors.

End of code in loop, makes Arduino read from beginning of void loop again. [REPEAT]

Development of Haptic suit



Tracing and cutting first layer of the suit, the one which has holes for the haptic pods



Attaching Synthetic Sports mesh to inside of first layer



Second Layer which holds the haptic motors against the body





Making Adjustable strap to that results in varying pressure of back Haptic Pods out of large elstic and Velcro.



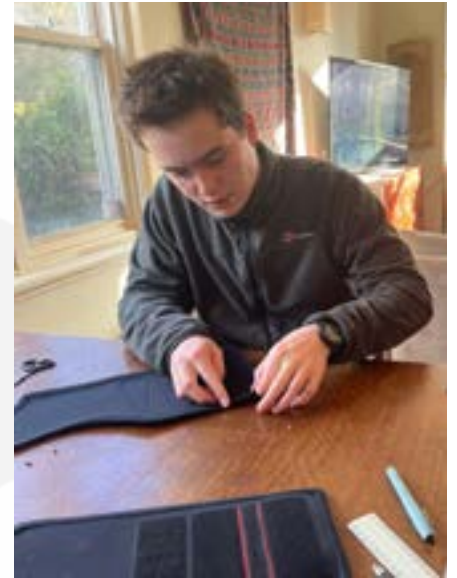
Putting third Layer of noprene on. This layer is the layer that protects the and cables directs them the the bottom of the suit where there is an enty point with Velcro which will hold the cables and make them un-pullable.



Cutting the front piece of neoprene into shape. The bottom of this piece of the suit has an arc from one side to the other. This is because once the fabric goes around a body it the will look flat and be more comfortable.



Velcro pieces added to front of the vest. The shoulder strap has a long piece of Velcro to allow for a large range tightening possibilities and the stomach adjustable pieces have a large surface area to allow for maximum adjustability. I have put flashes of red under the Velcro to add colour to the design.



Attaching the bias binding was a very time-consuming part of the construction of the suit but the results were very worth it. The bias binding definitely makes the design look like a finished product.

There was a piece of fabric of the shoulder which looked unfinished as I had to add another piece of neoprene to lengthen this part because neoprene did not come in sheets big enough. So to make this look like a deliberate piece i cut the attachment into a triangle and then added some red bias binding.



Bias Binding

There were pieces of the design that I could not sew with the machine so for this I would sew by hand. The main pieces of hand sewing that was required was to sew the binding. The binding was essential because the neoprene looked very unfinished with some uneven cuts and the most unfinished part was that when the edges were sewn, they splayed out and as neoprene is very difficult to sew there were uneven parts. The binding was essential but had to be hand sewn because 3 layers of neoprene did not fit through the sewing machine binding attachment. I did not want the stitching to be seen though, so I sewed one side of the binding backwards then flipped it over which hid the sewing on the outer side, and then I backstitched the other side which predominantly hid the thread.

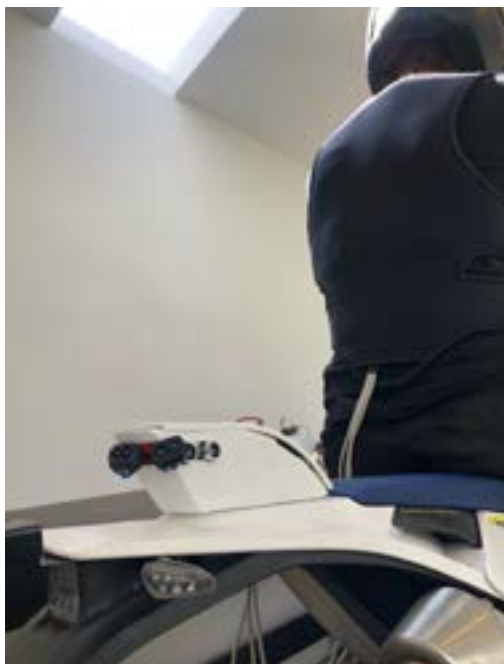




To finish the suit I sewed a zip on to the front. Unfortunately I could not hide the stitching of the zip through the bias binding but it is two straight lines of stitching so it looks neat.



Suit and Electronics all together



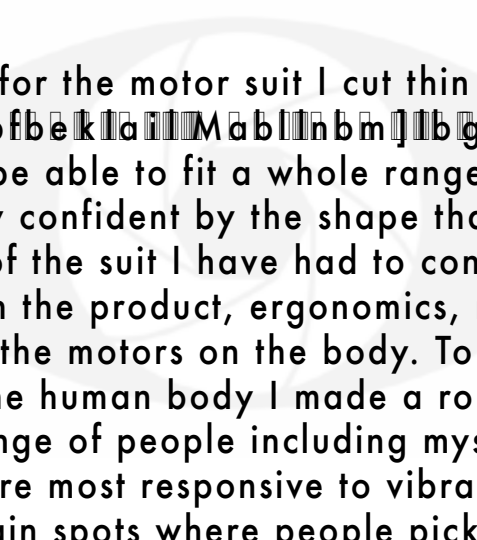
Feeling sight

Ongoing evaluation

Fabric

I am happy with the material choices I have made so far. I have chosen to use sport fabrics that a polyester as the polyester wicks away sweat and does not become incredibly heavy when moisture is absorbed. I have also discovered that neoprene is a great fabric because of the way it is ably to securely hold its shape whilst being comfortable and soft for the user. There is one draw back to neoprene that I have discovered and did not account for. This is that neoprene is a very difficult fabric for conventional sewing machines and industrial sewing machines to sew. The needle often does not fully penetrate the fabric which leaves the cotton without anything to attach itself around in the machine. But I did discover after research that sewing machines can have walking feet which help move the fabric without contorting it. This proved to be the best solution with it almost always catching after this addition.

Suit

To get a suitable shape for the motor suit I cut thin cardboard to the shape  be adjustable so it will be able to fit a whole range of people after cutting the cardboard I am very confident by the shape that has morphed together. In the development of the suit I have had to consider cable management, the ability to wash the product, ergonomics, making sure cables do not pull and position of the motors on the body. To find the perfect position for the motors on the human body I made a rough haptic pod prototype and tested on a range of people including myself, testing what parts of the back and side were most responsive to vibrations. I concluded that there was only a few main spots where people picked up the vibrations best and that is where I fixed the motors.

Haptic pod

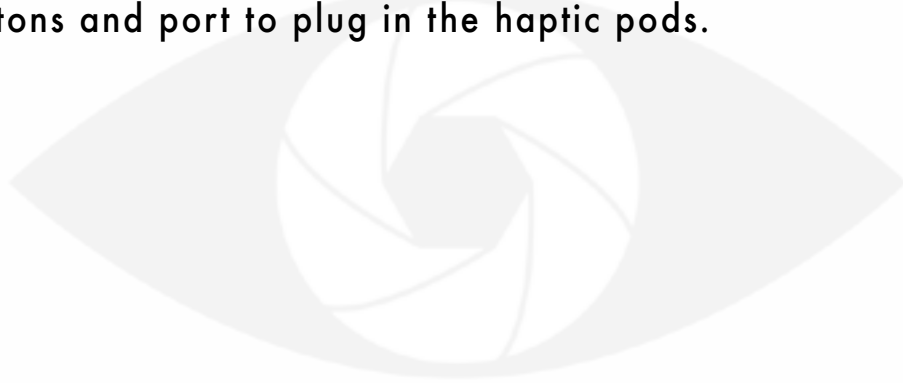
As I designed the haptic pods I had very few issues but I 3D printed multiple prototypes because I just really wanted to get the haptic pods to the correct size so it would be felt well by the user and would also be small enough that the user can differentiate between each boxes position. I also made sure the haptic motors fitted in the shell perfectly.

Electronics

HkbbgeerBmahnamBphneJnlkkllkkrrbmmakkgllhmmakhc -
ect. With using the raspberry pi I started to program a video processing program but this proved to be a very difficult ask of the raspberry pi. The raspberry pi could not process this information quickly enough for it to be applied in a safety application. After this unfortunate turnout I decided to use lidar (light detection and ranging) connected to an Arduino. This knew configuration of devices proved to be far quicker at processing and achieved the same outcome.

Fender Box

When coming up with a fender mount box to hold all the electronics I had to firstly consider the size of the electronics and make a box that they could fit in. Next I had to consider the shape of the fender and work out the basic dimensions I was working with for the fender. After measuring up both of these aspects I then designed the fender box in a way that looks aesthetically pleasing and also has the function of having an LCD screen and various control buttons and port to plug in the haptic pods.



Degree of difference

This conceptual suit design will be very different to any of the other designs that are on the market. The Classon helmet is the closest design that is on the market similar to my concept suit design. But this helmet is designed for bicycle riders to give the bicycle functions that all road registers motorcycles already have. This device unfortunately has no way of telling the user what is going on in the lanes behind them.

MaAkJeBaamOKInbmblbfbekhgmmhmbablkhcmmneInbmInm the purpose of the suit is for gaming. This suit cannot be directly repurposed to be used in the application that I need it to be. This suit does not have haptic motors positioned in the correct positions to transfer information taken in by the sensors to the rider.

The suit that helps deaf people with hearing the world is a great innovation and is a similar concept to the suit I will create. This suit remaps the hkkkkbgmhbgmbmmkgbbgmaInbmffhmkppnggImkglem that into sound. My suit is going to take information about the position of vehicles on the road and transfer that to the rider using haptics.

Rear vision mirrors are a good design because humans are very good at identifying movement with their eyes. But the limitation with mirrors on motorcycles is that it does not have blind spot detection or the ability to see directly behind. This is where blind spot detectors come in for bikes but they are expensive and often disregarded as a good thing to put on a bike.

Project Evaluation



Aanlysis and evaluation of functional and aesthetic aspects of the design

Functional

Function	Positive	Negative	Evaluation
Will provide rider with information about vehicles around them	This suit provides tells the rider where vehicles are around them when they are on the roads.	The suit does tell the rider what type of vehicle it is.	This is the main purpose of the suit and the suit provides good information to the rider. This has been a success.
Will Provide user with information about their surroundings and any other analytics	Because of the build of the suit there is every possibility that with additions the to software this is easily possible.	Unfortunately due to lack of time this is not a feature in the suit.	This suit has been made to be able to facilitate functions such as this but is not a current feature.
Battery pack and processing unit able to be stored safely	All the electronics are safely stored in a box that sits on the back of bike.	This box is currently 3D printed plastic. It would be good if it could be made of a stronger material in future.	The Project functions well in this aspect.
Has battery life that will last at least a full day of activity or have changeable batteries.	The electronics currently use disposable batteries that will last 2-3 days of continuous use.	Unfortunately this project does not have rechargeable lithium batteries because they were difficult to source during covid.	The battery situation is not bad but a device such as this should be able to be re-charged.
Connect to power source from the Motorcycle	Connecting to the motorcycle battery would be ideal. And with the use of DC step down chips this is possible.	I did not have the time to do this. And I have never done something like that before.	This would have been a good way to power the project but for demonstration of concept portable batteries are ideal.
Be able to measure corners and how fast someone should go into them	The suit can facilitate this feature.	Was not able to bring this feature into the device. Due to not using computer vision.	If computer vision is used in future prototypes this feature is easily possible.

Aesthetics/Ergonomics

Aesthetics/ Ergonomics	Positive	Negative	Evaluation
User is able to feel the motors even with an engine running	This suit has succeeded very well with this intended feature.	The user has to ensure they use the adjustable straps fit the suit correctly.	The user will be able to feel the vibrations very well as long as the suit fits properly.
Discrete in clothing	This suit design has actually surprised me with how discreet it is. When Motorcycle clothing is worn the suit is basically non-visible.	N/A	The design has succeeded at the given criteria for aesthetics.
Does not create bulges in fabric	This suit hides the haptic motors very well.	There are slight bulges in the back but nothing that ruins the aesthetics of the suit.	The suit has succeeded at this criteria making the haptic pods a feature.
Comfortable	This suit is very comfortable because of using the solid but moveable features that neoprene provides	N/A	The suit has succeeded well at this aesthetic criteria.
Fits body and sits in same place each use	Because of the ability to adjust the suit and keep it in that position even when removed this works very well.	N/A	This ergonomic criteria has been met and past because of the suits ability to only need to be adjusted once.
Can be adjusted for different users	Because of the two separate front and back pieces this suit can be fully customised to fit a wide range of people.	If someone is a lot smaller or a lot bigger this suit will need to be remade to fit them. But for a range of average adult sizes it fits well.	This suit has met this ergonomic criteria well.

Final evaluation with respect to the projects impact on the individual, society and the environment

Individual

This product will affect individuals who ride motorcycles on the road by providing another layer of safety which does not inhibit their performance. The product that I have made is both a textile, electronic and 3D design. These components are the types of things which are in many products in the current world that affect us most. The textiles component of this project affects individuals by making a safety technology work with the un-uniform shape of the human body. The electronic components and programs built to operate this safety technology are the things that will save individuals lives and make them feel safer when riding on the road. And the 3D design of the Fender Box has been made to look aerodynamic and functional in hopes to not deter users from using a safety technology such as this.

Unfortunately, a technology such as this will not be cheap for an individual to buy due to the price of electronic components and textile materials. This leaves people who ride motorcycles because they cannot afford to run a car, less safe on the roads. This is not what I want out of this project. I want a product that many people can afford and use no matter their income. This is possible though, by bulk buying components and developing this technology even further the price to manufacture one of these devices would be drastically reduced.

Society

If this haptic suit were to be used by many motorcyclists on the roads, it would impact the whole society by reducing accidents and deaths on the roads. Accidents and deaths that occur on roads causes great distress for the people and families involved. But not only that, accidents on the roads of Australia cost societies economy \$22.2 billion dollars with each fatality costing \$4.3 million dollars. This statistic is not a pleasant statistic because bmanmllhilellmknlfgjleboollhpgmhlhghmkröen%lmmabllmmbll - tic has to be looked at because the government could be putting this money towards other things such as their foreign aid budget which helps people in suffering countries, at the moment the Australian government will spend an estimated total of \$4.3 billion dollars helping other countries. Imagine the possibilities with even half of the accidents budget. This suit alone is not going to help reduce the number of accidents on the road but if society starts using and creating more safety devices such as this suit it will save lives and be able to provide better funding underfunded aspects of society.

Environmental

This suit has not been made to be environmentally as the first priority in the current prototype. This current suit is a proof-of-concept design. In future, designs can be made from biodegradable materials which will act as the housing for the fender box and haptic pods. There also are biodegradable rubber substances which I believe can be used to produce more ecofriendly neoprene. But most of all the most eco-friendly part of this device is that it promotes the use of motorcycles as a legitimate safer option to driving cars. Motorcycles are the most mainstream ecofriendly form of transportation using combustible fuel. If more motorcycles were used as modes of commute this would greatly affect the emissions of vehicles in society.



Relationship of the final product, system or environment to the proposal

In my proposal for this project I proposed that I develop a system that will allow motorcyclists to know exactly where other vehicles are around them to increase safety and reduce accidents. From this I aimed to help motorcyclists know exactly where they are in space and where other vehicles and dangerous obstacles are without needing to see them. This will help riders avoid collisions and increase safety on road and track.

This project has been successful in respect to my initial intentions I have made a suit that allows motorcycle riders know where other vehicles on the road are in their frame of reference. However I feel the resultant project that I will submit is only showing some of the features a device like this could have. I feel that with communication that is transferred through feeling in particular haptics has so many potentials. For example with this technology the analytics of a vehicle could be transferred to the operator through haptics this would make riding/driving safer and could make racing even closer and more intricate. This is not the only use I see for technology such as this. I think that if this suit got repurposed and had some adjustments with the use of computer vision, this technology could help blind people physically navigate life with a greater ease. With research I had done I came across David Eagleman an American neuroscientist who had done research on how the human brain is able to remap complicated pathways to allow humans to easily recognise new patterns. He demonstrated this by building a device that also tapped into the sense of feeling for deaf people. And the device translated sound into feelings and quite quickly deaf people were able to understand language and various sounds.

As my final evaluation of this project, I believe this product is a good demonstration of what haptics can achieve to help keep motorcyclists safer on the roads. I also feel as though this project is incomplete because there are so many more possibilities to with motorcycles alone that can be achieved given more time. I had issues with integrating so many components into one electronic device. In future designs I would look at creating my own specific electronic boards already with these components integrated. This would result in less things that could go wrong and it would make the physical electronics volume far smaller with the ability to make more stylish seamless Electronic box designs.

Final Product to End User

This product is still in the development phase for the electronics, because I want develop it further by making a custom PCB board which made specifically for the suit. I also like the idea adding more features that are far beyond the aim of this project like vehicle analytics. I also think this device could be made wireless which would make using it incredibly easy.

The final product out of this project though is a usable suit which relays real time information collected by an Arduino using haptic pods. This information is collected from 3 lidar sensors and 2 ultrasonic sensors. With being able to have riders use this technology for 3 or more years, which is the dedicated time for statistical patterns to form, it is impossible to say one way or the other what the effect of this suit is. But if it does succeed at its intended purpose this is a good preventative safety device.



Acknowledgements

- Sharron Hurry, for her time and advice
- My Parents who helped fund this project
- My Design and Technology teacher who helped with guidance and support

